Production balances and a normalized production diagnostic for invariant measures of the stochastic gCLM ($a = -2$): Galerkin-level identities and a large-viscosity limit

Rifat Jumagulov
jum.rifm@gmail.com

Status labels used throughout: ***proved*** *(Appendices B and C, with symbolic certification as an independent layer)* · ***measured*** *(spectral solver at controlled stationarity)* · ***conditional*** *(proved modulo a named input)* · ***open****. Verification details: Appendix A.*

Abstract

We study invariant measures of the stochastically forced generalized Constantin–Lax–Majda–De Gregorio equation at $a = -2$, the enstrophy-conserving case (Fujita–Fukuizumi–Sakajo). Applying the stationarity identity $E_\mu[\mathcal{L}F] = 0$ to production functionals, we obtain, at each fixed Galerkin truncation, a family of exact balances. The central object is a *production balance* for $\mathcal{P}_1 = \int (H\omega)\omega_x^2$, carried exactly by the *projected* inertial pairing $I_M = \langle P_M N, P_M D\mathcal{P}_1 \rangle$ — not the full functional I, from which it differs by an explicit Galerkin defect R_M that vanishes exactly for band-limited fields ($2K \le M$); it defines a **normalized production diagnostic** $\theta = 1 - E_\mu[N_1]/E_\mu[W_1]$. We establish three properties of θ. (i) It is *scale-blind*: dilation-invariant while spectral localization is not, so no bound factoring through θ alone controls the low-mode dissipation. (ii) It is *grouping-dependent* — only $E_\mu[I]$, the exact balance $E_\mu[I_M] = -\mathcal{C}_1$, and the spectral flux are convention-free — so its numerical value is a diagnostic, not a physical cascade magnitude. (iii) **(Theorem F, the main result)** at large viscosity $\theta \to \theta_G(\text{shape}) + O(\varepsilon)$ for $M \ge k_f$, and $\theta_M \to \theta_G$ with the balance coefficient positive for $M \ge 2k_f$, where $\varepsilon = \sqrt{B_0}\,\nu^{-3/2}$; this holds for *every* invariant measure of the Galerkin system, with the exact Gaussian value $\theta_G(\text{flat band}\,[1,4]) = 2017/2484$ (for the stated W_1-grouping). A stationarity- and convergence-controlled pseudo-spectral campaign measures $\theta \in [0.77, 0.81]$ across $\nu \in [0.005, 0.2]$. **Scope.** All identities and Theorem F are theorems at each fixed Galerkin truncation; the cubic/quintic production balances are *formal* for the untruncated SPDE (an open generator-domain justification), whereas the quadratic enstrophy balance and the cylindrical spectral-flux law (§3) are more readily infinite-dimensional. Every algebraic identity is proved analytically and independently certified symbolically.

Keywords: generalized Constantin–Lax–Majda equation; De Gregorio model; stochastic PDE; invariant measure; enstrophy cascade; flux law; production balance; production diagnostic.
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1 Introduction

The generalized Constantin–Lax–Majda–De Gregorio (gCLM) equation $\omega_t + au\omega_x - u_x\omega = \nu\omega_{xx}$, $u_x = H\omega$, is the canonical 1D caricature of 3D vortex stretching under advection: the term $u_x\omega$ stretches, the term $au\omega_x$ transports, and the balance between them (governed by a) is a sharp, analytically tractable model of the competition that controls regularity in the Euler and Navier–Stokes equations (Constantin–Lax–Majda 1985 [3]; De Gregorio 1990 [4]; Okamoto–Sakajo–Wunsch 2008 [11]; Elgindi–Jeong [6]; Chen–Hou–Huang [2]). At $a = -2$ the nonlinearity conserves the enstrophy $\frac{1}{2}\int \omega^2$, and the stochastically forced model is the natural setting for the **anomalous enstrophy cascade** — the 1D analog of the 2D enstrophy cascade. The randomly forced model has been studied for over a decade: Matsumoto–Sakajo demonstrated the anomalous enstrophy cascade, its spectrum and intermittency numerically [9], connected the forced turbulent state to the inviscid singularity across the gCLM family [10], and Tsuji–Sakajo established statistical laws for the randomly forced model as a random PDE [12]. Its rigorous invariant-measure (ergodic) theory was given by Fujita–Fukuizumi–Sakajo [7], who proved existence of an invariant measure and, at large viscosity, uniqueness and exponential mixing.

This paper develops a *production-balance* theory for the invariant measures of (SG). The construction is elementary and intrinsic to the model: for every invariant measure the stationarity identity $E_\mu[\mathcal{L}F] = 0$ ($\mathcal{L}$ the Itô generator) applied to a *production functional* F yields an exact **production balance** relating a sign-definite "production of production" W to a nonlocal "coherence" term N, whose ratio is a dimensionless **normalized production diagnostic** $\theta = 1 - E_\mu[N]/E_\mu[W]$ (the neutral name is used because θ is grouping-dependent, §4–§5). At a fixed truncation the exact balance is carried by the *projected* inertial pairing, a distinction made precise in §4. The same generator-identity strategy underlies invariant-measure analyses of stochastically forced fluid models more broadly; here it is realized in a setting where the whole structure is (a) exactly and cleanly closed, (b) CAS-certifiable term by term, and (c) measurable in isolation on the accompanying pseudo-spectral solver. The scalar θ is measured to be positive and ν-stable ($\theta \approx 0.78$ at small ν, rising to its Gaussian limit $\theta_G \approx 0.812$ as $\nu \to \infty$, §4); whether a uniform floor $\theta \geq \theta_0 > 0$ can be *proved* is the paper's central open problem (§6), which we show is not reachable by soft-algebraic means at small ν and reduces to a named scale-resolved closure input — while at large ν positivity becomes a concrete perturbative target ($\theta \to \theta_G$, §6 paragraph (S6)).

Contributions. The distinctive object of this paper is the *production-balance / production-diagnostic* construction (item 2 below): a CAS-certified stationary balance whose scalar θ is measurable in isolation on the accompanying pseudo-spectral solver. The enstrophy-flux law and its UV-concentration reformulation (item 1) are, as we note in §3, *known in spirit* (Bedrossian–Coti Zelati–Punshon-Smith–Weber [1]; Dudley [5]; Liu–Lu–Wang [8]); we include them not for novelty but because A1$''$ is the exact *input* that the production-diagnostic analysis is built on.

1. An exact **enstrophy-flux law** for the stationary measure (Theorems A0–A2, §3), with the inertial-range constant $\frac{1}{2}B_0$ in the inviscid limit (conditional on the scale-resolved input (†-weak) of §6, a strict weakening of a uniform bound), and the exact palinstrophy production balance. *(Known in spirit; here realized cleanly on the certified $a = -2$ identities as the input to item 2.)*

2. The **production balance** for the palinstrophy production functional $\mathcal{P}_1 = \int (H\omega)\omega_x^2$ (§4), stated exactly at the Galerkin level through the projected pairing I_M (with the Galerkin defect R_M identified — exactly vanishing under a spectral-support/band-limitation condition ($2K \leq M$, Lemma 4.1) — and measured), defining the normalized production diagnostic θ; every term proved (Appendix C) and CAS-certified.

3. A **reduction** of a mean-coherence bound to a lower bound on a single destruction, a **conditional** coherence bound under a diagnostic-floor hypothesis, and a **reverse-Markov density** bound on negative inertial-generation episodes (§5).

4. The **obstruction analysis** (§6): a hypothesis-free reformulation of the flux law as UV-concentration of ρ_ν, a strict weakening of its open input to (†-weak), a hereditary-cancellation lemma, and a **scale-blindness lemma** (Lemma 6.1): θ is invariant under mode dilation while spectral localization is not, so no bound factoring through the scalar θ alone can close (†-weak); the uniform (small-ν) floor $\theta_0 > 0$ remains open.

5. **Theorem F** (§6): at the Galerkin level ($M \geq k_f$; $M \geq 2k_f$ for the θ_M half), $\theta(\mu_\nu)$, $\theta_M(\mu_\nu) = \theta_G(\text{shape}) + O(\sqrt{B_0}\,\nu^{-3/2})$ for every invariant measure — in particular $\mathcal{C}_1 < 0$, positivity of the balance coefficient, *proved* at large viscosity, with the exact (W_1-grouping-relative, §5) limit $\theta_G(\text{flat band}[1,4]) = 2017/2484$.

Main theorem, stated precisely (Theorem F, §6). This is the paper's principal new result; we state it here so the rest of the paper reads as the route to it. Write the small parameter

$$\varepsilon := \sqrt{B_0}\,\nu^{-3/2}, \tag{1}$$

the size of the non-Gaussian correction to the $\nu \to \infty$ Ornstein–Uhlenbeck law. Fix a homogeneous forcing band $[1, k_f]$ and a Galerkin cutoff M. Then for *every* invariant measure μ_ν of the Galerkin system there are a constant $C = C(M, \text{shape})$ and thresholds $\nu_0 \leq \nu_1$ (each depending on M, B_0, shape) with

$$\big|\theta(\mu_\nu) - \theta_G(\text{shape})\big| \leq C\,\varepsilon \;\;(M \geq k_f), \qquad \big|\theta_M(\mu_\nu) - \theta_G(\text{shape})\big| \leq C\,\varepsilon \;\;(M \geq 2k_f), \tag{2}$$

for all $\nu \geq \nu_0$, where $\theta_G(\text{shape}) = E_{G_\nu}[I]/E_{G_\nu}[W_1]$ on the degenerate Gaussian $G_\nu = \mathcal{N}(0, \text{diag}(b_k^2/(2\nu k^2)))$. In particular, for $\nu \geq \nu_1$ and $M \geq 2k_f$, $\mathcal{C}_1 < 0$ and $\theta_M > 0$ hold as *theorems*. Three features are essential and are *not* hidden: (i) C and the thresholds grow with M — the bound is not uniform in the cutoff, so it does not by itself pass to the untruncated SPDE; (ii) the ratio is controlled through the strictly positive Gaussian denominator $E_{G_\nu}[W_1] = c_W\, s^4 > 0$ (with $s_k^2 \propto b_k^2/k^2$); (iii) G_ν is genuinely degenerate on the unforced modes, which carry no variance. The full statement, the itemised constant-dependence, and the proof are in §6 and Appendix B; for the flat band $[1,4]$, $\theta_G = 2017/2484$ (for the stated W_1-grouping, §5).

All numerics are on the accompanying pseudo-spectral solver; every algebraic identity is proved analytically for general fields (Appendix C) and certified symbolically as an independent layer (residual $\equiv 0$). Every pointwise/threshold surrogate that fails is recorded, and every "for all invariant μ" statement is stated without assuming the (unknown at small ν) uniqueness.

2 Setting

On the torus $x \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, real mean-zero vorticity ω,

$$d\omega = \big(-au\omega_x + u_x\omega + \nu\omega_{xx}\big)\,dt + dW, \qquad u_x = H\omega, \qquad a = -2. \tag{SG}$$

$(H\omega)\hat{}(k) = -i\,\text{sgn}(k)\,\hat{\omega}(k)$, $\text{sgn}(0) = 0$; $u = -\Lambda^{-1}\omega$ so $u_x = H\omega$; write $v := u_x = H\omega$ (the stretching rate). $W = \sum_{k\in\mathbb{Z}^*} b_k\beta_k e_k$ is additive, white-in-time, $\{\beta_k\}$ independent, real ON basis $\{e_k\}$ ($\int e_k^2 = 1$; concretely $e = \cos(kx)/\sqrt{\pi}$ and $\sin(kx)/\sqrt{\pi}$ per $k \geq 1$, the index set $\mathbb{Z}^*$ enumerating the two members of each pair as $\pm k$); moment sums $B_0 = \sum_k b_k^2$, $B_2 = \sum_k b_k^2 k^2$; normalization $E\|\xi(t)\|_{L^2}^2/t = B_0$. The normalized *shape* of the forcing is the profile $\bar{b}_k := b_k/\sqrt{B_0}$ ($\sum_k \bar{b}_k^2 = 1$): an overall amplitude rescaling is absorbed into B_0, whereas the shape $\{\bar{b}_k\}$ is what enters θ_G(shape) and the constants C, ν_0, ν_1 of Theorem F (which therefore depend on the shape, not on the amplitude, at fixed M). "Homogeneous" noise means $b_k = f(|k|)$ — equal amplitudes on the cos / sin pair of each $|k|$, i.e. a noise field whose law is invariant under spatial translation — supported on a band $1 \le |k| \le k_f$. The *flat* band $[1, k_f]$ is the special case $b_k \equiv b$ (equal amplitude on every forced $|k|$), for which $B_0 = 2k_f b^2$; the campaign uses the flat band $[1, 4]$ with $B_0 = 0.16$, and the exact Gaussian value of §4 is computed for it. μ = any invariant probability measure of (SG) (FFS: exists; unique + mixing at large ν).

Functionals. enstrophy $\mathcal{E} = \frac{1}{2}\int \omega^2$; palinstrophy $\mathcal{E}_1 = \frac{1}{2}\int \omega_x^2$; enstrophy dissipation $\nu \int \omega_x^2$; palinstrophy production $\mathcal{P}_1 = \int (H\omega)\omega_x^2$; enstrophy flux $\Pi(K) = -\langle P_{<K}\omega, P_{<K}N[\omega]\rangle$ ($P_{<K}$ = low-pass $|k| < K$, $N[\omega] = -au\omega_x + v\omega$). **Low/high flux decomposition.** Writing $\omega = \omega_L + \omega_H$ with $\omega_L = P_{<K}\omega$, $\omega_H = (\mathrm{Id} - P_{<K})\omega$ and expanding the quadratic N through its symmetric bilinear form $N[\cdot,\cdot]$, the flux splits as $\Pi(K) = \Pi_{LL} + \Pi_{LH} + \Pi_{HL} + \Pi_{HH}$, $\Pi_{XY} := -\langle P_{<K}\omega,\ P_{<K}N[\omega_X, \omega_Y]\rangle$; the *HH share* $\Pi_{HH}/\Pi(K)$ is the fraction of the flux carried by high–high interactions (columns in Table 1; convergence in Table 5).

Itô generator. For a functional F, $\mathcal{L}F = \langle b, DF\rangle + \frac{1}{2}\mathrm{Tr}(QD^2F)$, $b = -au\omega_x + v\omega + \nu\omega_{xx}$, $Qe_k = b_k^2 e_k$; and $E_\mu[\mathcal{L}F] = 0$ for every invariant μ and every F in the (polynomial) domain. *Domain remark.* At any fixed Galerkin truncation the identity $E_\mu[\mathcal{L}F] = 0$ is *rigorous* for every polynomial F and every invariant μ — the Lyapunov/moment argument is displayed as Lemma B.2 in Appendix B. The solver used throughout evolves the genuine Galerkin system on E_M, $M = \lfloor (N-1)/3 \rfloor$ (strict 2/3 rule; both the nonlinear drift and the state are projected to E_M, so with band-limited noise the state remains in E_M identically), with the nonlinear drift advanced by an explicit first-order step in time: the simulated Markov chain is a time-discretization whose stationary law approximates that of the continuous Galerkin SDE. The scheme is *formally first order* in dt; we do not claim an a priori weak-error order for its invariant measure, and the resulting bias is controlled *empirically* by step-halving (Appendix A). Measured statements are therefore *estimates of the continuous Galerkin system's stationary averages* — not exact evaluations. *Caution (test functions), and a hierarchy of difficulty.* An identity $E_\mu[\mathcal{L}F] = 0$ is exact at a truncation with the *projected* gradient $P_M DF$. The functionals used here are *not* all of equal difficulty for the untruncated SPDE, and we do not lump them under one blanket caveat: (i) the spectral flux law (Theorem A1) uses the *cylindrical* functional $F_K = \frac{1}{2}\|P_{<K}\omega\|^2$, which depends on finitely many modes and whose gradient $P_{<K}\omega$ is band-limited, so its stationary identity is the most directly justified at the SPDE level (standard cylindrical-function generator theory); (ii) the enstrophy balance (Theorem A0) uses the quadratic Lyapunov functional $\frac{1}{2}\|\omega\|^2$, standard but not cylindrical; (iii) the production balance of §4 uses the cubic $\mathcal{P}_1$, whose gradient $D\mathcal{P}_1$ leaves E_M and carries a quantified projection defect (§4). For (iii) — the cubic/quintic functionals — we do *not* claim the (nontrivial, open) infinite-dimensional generator-domain justification for the untruncated SPDE; this is the explicit caveat that separates the genuinely finite-dimensional results (all of §4–§6, including Theorem F) from the more readily infinite-dimensional §3 flux law. **Convention:** throughout §3–§6, "for every invariant measure" is to be read rigorously at any fixed Galerkin truncation (where all identities and Theorem F are theorems), and formally — exactly at each truncation, modulo the stated domain caveat — for the untruncated SPDE.

3 The enstrophy-flux law

All of §3 is proved: full arguments in Appendix C (integration by parts and the Hilbert-transform calculus, valid for general smooth fields and holding at every Galerkin truncation), with independent symbolic and spectral certification recorded in Appendix A. The inviscid-limit constancy (FL∞) *is conditional on the named input* (†).

Theorem A0 (Enstrophy balance)**.** *For every invariant measure μ (rigorous at every fixed Galerkin truncation; formal for (SG), §2),*

$$\nu\, E_\mu\big[\textstyle\int \omega_x^2\big] = \tfrac{1}{2}B_0 \qquad \textit{(exact, all } \nu\textit{).} \tag{3}$$

Proof: Appendix C (C.1–C.2). The nonlinear contribution $(1 + a/2)E_\mu[\int v\omega^2]$ vanishes at $a = -2$ (advective cancellation, Lemma C.1; certified to residual 10^{-19}). *Consequence:* enstrophy dissipation is pinned at $\frac{1}{2}B_0$ independent of ν (anomalous dissipation of the conserved quantity); $E_\mu[\mathcal{E}_1] = B_0/(4\nu) \to \infty$ as $\nu \to 0$. Solver: $\nu E[\int \omega_x^2]/(\frac{1}{2}B_0) = 1.006$.

Theorem A1 (Spectral enstrophy-flux identity)**.** *For every invariant μ (Galerkin-rigorous; formal for (SG), §2) and every K,*

$$E_\mu[\Pi(K)] = \tfrac{1}{2}B_0^{<K} - \nu\, E_\mu\|P_{<K}\omega_x\|^2, \qquad B_0^{<K} := \sum_{|k|<K} b_k^2 \tag{4}$$

($\Pi(K)$ *is the random flux functional of §2; the identity holds for its stationary mean).*

Proof: Appendix C (C.2).

Below the forced band $\Pi(K) \approx 0$; across the inertial range $\Pi(K) = \frac{1}{2}B_0 - \nu E\|P_{<K}\omega_x\|^2$; and $\Pi(K) \to 0$ in the dissipation range (recovering A0 as $K \to \infty$).

Corollary A1$''$ (Anomalous-cascade flux law as UV-concentration; unconditional given Theorem A1, at fixed truncation)**.** *Define the* ***enstrophy-dissipation measure*** *on $k \in \mathbb{Z}^*$, $\rho_\nu(\{k\}) := \nu k^2 E_\mu|\hat\omega(k)|^2$. By (L0) its total mass is $\rho_\nu(\mathbb{Z}^*) = \nu E_\mu\|\omega_x\|^2 = \frac{1}{2}B_0$,* ***$\nu$-independent (the same total mass at every ν)****. Then for each fixed inertial-range K ($K > k_f$), from the identity (FL) with no estimate:*

$$\lim_{\nu\to 0}\Pi(K) = \tfrac{1}{2}B_0 \quad\iff\quad \rho_\nu(\{|k| < K\}) \to 0 \quad\iff\quad \rho_\nu \textit{ UV-concentrates past } K. \tag{5}$$

Hence the inertial-range flux law (FL∞) *at every K is* **equivalent***, given only the certified identities, to the statement that the fixed budget $\frac{1}{2}B_0$ of enstrophy dissipation migrates to arbitrarily small scales as $\nu \to 0$.* $\square$

Scope (order of limits), stated precisely. The inviscid-limit reading requires a joint limit, which we fix once. Fix $K > k_f$. Take sequences $\nu_n \downarrow 0$ and cutoffs $M_n \to \infty$ with $M_n > K$, and for each n let μ_{ν_n,M_n} be *any* invariant measure of the M_n-truncated system. Then, from the certified identity (FL) alone,

$$\Pi(K) \longrightarrow \tfrac{1}{2}B_0 \qquad\iff\qquad \rho_{\nu_n,M_n}(\{|k| < K\}) \longrightarrow 0, \tag{6}$$

i.e. the flux law at K is *equivalent* to the drain of the low-mode dissipation mass. At a *fixed* M this content is empty for $K > M$ (there $\Pi(K) \equiv 0$, since the low-pass set then holds the entire dissipation measure); the rigorous content at fixed M is the drain of $\rho_\nu(\{|k| < K\})$ for each

$k_f < K \leq M$. Thus "for every fixed K" quantifies over K *after* the truncation — never at a single fixed M — or formally for the untruncated SPDE (§2 convention).

This is the honest, hypothesis-free form of the flux law: (FL∞) **is** the UV-concentration of ρ_ν, definitionally. It is a Theorem only in the trivial sense ($x \to c \iff c - x \to 0$ applied to the certified identity of Theorem A1); all the analytic content resides in Theorem A1, which is why we record it as a corollary rather than a separate result.

Relation to prior work (no conceptual novelty claimed for A1$''$). The flux-law $\iff$ spectral-concentration equivalence is not new in spirit: sufficient conditions for stochastically-forced dual-cascade flux laws were given by Bedrossian–Coti Zelati–Punshon-Smith–Weber (2019) [1], sharpened to a necessary-and-sufficient form by Dudley (2023) [5], with the analogous SQG statements in Liu–Lu–Wang (2026) [8]. Our contribution is only (i) the clean $a = -2$ realization on the model's own certified identities, and (ii) its role as the *input* to the production-diagnostic analysis of §4, which is the distinctive object of this paper.
The remaining open content is exactly whether $\rho_\nu(\{|k| < K\}) \to 0$, equivalently

(†-weak) $\quad \nu E_\mu \|P_{<K}\omega_x\|^2 \to 0$, i.e. $E_\mu \|P_{<K}\omega\|^2 = o(1/\nu)$,

a strict weakening of the naive uniform bound (†): $\sup_\nu E_\mu \|P_{<K}\omega\|^2 < \infty$: Poincaré from (L0) already gives $E_\mu \|P_{<K}\omega\|^2 = O(1/\nu)$, so (†-weak) needs only an $o(1)$ (e.g. a single logarithm) improvement, not boundedness. (†-weak) is the **central open target** (§6). *(Corollary A1$''$ and (†-weak) are self-contained here, proved directly from (L0)/(FL); §6 gives the consolidated verdict on the attempts to close (†-weak) and $\theta \geq \theta_0 > 0$.)*

Measured (stationarity-controlled campaign of §4; forced band $[1,4]$, $B_0 = 0.16$, split-half $z < 2$ on the observables, two seeds per point).

(i) $\nu E[\int \omega_x^2]/(\frac{1}{2}B_0) \in [0.977, 1.030]$ (seed-pooled per ν) across $\nu \in [0.005, 0.2]$ — (L0) holds uniformly in ν.

(ii) The palinstrophy scaling $E[\mathcal{E}_1] = C\nu^p$ fits $p = -1.002$, $R^2 = 0.9999$, $C = 0.0400$ vs $B_0/4 = 0.0400$ — the enstrophy-production law $E[\mathcal{E}_1] = B_0/(4\nu)$ recovered exactly.

(iii) *(N.B. by (L0) the relation $E[\mathcal{E}_1] = B_0/(4\nu)$ is an exact identity, so the regression in (ii) is a solver/stationarity check, not independent evidence of the cascade; the cascade itself was demonstrated in this model class by Matsumoto–Sakajo [9, 10].)* Via the exact identity of Theorem A1 ($\Pi(K) = \frac{1}{2}B_0 - \rho_\nu(\{|k| < K\})$ for $K > k_f$), the stationary campaign gives the flux directly (Figure 1): at $\nu = 0.005$ it rises to a **broad maximum of** 0.87 **at** $K = 5$, then declines monotonically ($0.872 \to 0.834 \to 0.757 \to 0.600$ over $K = 5, 6, 8, 12$), against $\Pi(5)/(\frac{1}{2}B_0) = 0.06$ at $\nu = 0.1$ — the flux maximum *approaching* $\frac{1}{2}B_0$ as $\nu \to 0$, then decaying to 0 in the dissipation range ($\Pi \to 0$ as $K \to \infty$, recovering the $a = -2$ full-field cancellation). **We stress that this is not a resolved inertial-range plateau**: with forcing on $[1,4]$ and the maximum at $K = 5$ there is under one octave of scale separation, so the curve is a rounded peak, not a flat range. The exact equivalence (Corollary A1$''$) is what carries (FL∞); the numerics only illustrate the emerging maximum, they do not confirm a plateau.

An earlier coarse exploratory sweep is retained only for the a-dependence reported next; its recorded windows were not demonstrably at stationarity (split-half z up to 13–28, Appendix A), so we quote no error bars from it and have re-sourced every quantitative flux and scaling statement above to the stationary campaign of §4.

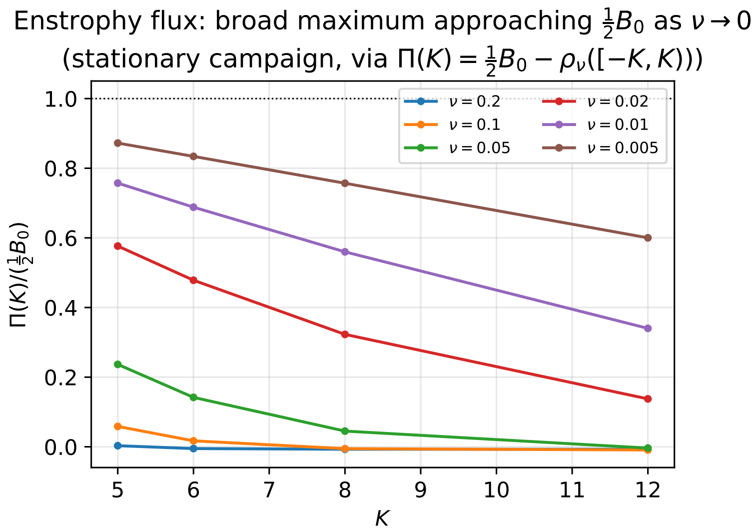


Figure 1: Enstrophy flux $\Pi(K)/(\frac{1}{2}B_0)$ from the stationarity-controlled campaign (§4), computed through the exact identity $\Pi(K) = \frac{1}{2}B_0 - \rho_\nu(\{|k| < K\})$ (Theorem A1, $K > k_f$): a broad maximum near $K = 5$ that grows and approaches $\frac{1}{2}B_0$ as $\nu \to 0$, then decays in the dissipation range. This is *not* a resolved inertial-range plateau (under one octave of scale separation between the forcing band $[1, 4]$ and the maximum). Statistical uncertainties (per-seed jackknife on ρ_ν, two seeds per point) are $\lesssim$ marker size; see Table 1. Data: `anc/p2_results_v18.jsonl`.

Bridge to the deterministic blow-up literature (qualitative). The coarse a-sweep ($\nu = 0.03$) shows the mean palinstrophy production **rising monotonically as advection weakens** ($E[\mathcal{P}_1]$ roughly tripling from $a = -2$ to $a = -0.5$), with the run **blowing up at** $a = 0$ (the CLM point, where finite-time singularity is expected). Being drawn from the exploratory sweep, this is a qualitative trend — we quote no uncertainties. It is *consistent with*, in the stationary stochastic setting, the deterministic principle that stronger advection depletes stretching and prevents singularity (Elgindi–Jeong [6]; Chen–Hou–Huang [2]), and with the forced-turbulence-vs-singularity study of the gCLM family by Matsumoto–Sakajo [10] — the $a = -2$ anomalous- cascade state is the maximally advection-balanced member of the family. *(The $a = 0$ run is flagged under-resolved ($z =$nan, Appendix A), so the blow-up there is consistent with the singularity but not by itself evidence for it; the monotone trend at $a \in [-2, -0.5]$ is the substantive observation.)*

Theorem A2 (Palinstrophy production balance)**.** *For every invariant μ (Galerkin-rigorous; formal for (SG), §2),*

$$\nu\, E_\mu\big[\textstyle\int \omega_{xx}^2\big] = E_\mu[\Pi_1] + \tfrac{1}{2}B_2, \qquad \Pi_1 = 2\textstyle\int (H\omega)\omega_x^2 + \int (H\omega_x)\,\omega\,\omega_x, \tag{7}$$

with the secondary term $\int (H\omega_x)\omega\omega_x = -\frac{1}{2}\int (H\omega_{xx})\omega^2$.

Proof: Appendix C (C.3); certified to residual 10^{-15} *(Appendix A).* Super-dissipation balances the nonlinear palinstrophy production plus injection $\frac{1}{2}B_2$; $E_\mu[\Pi_1]$ is the mean nonlinear contribution to the palinstrophy balance (its sign is not claimed a priori; measured positive, Appendix A).

4 The production balance and the normalized production diagnostic

Role of $\mathcal{P}_1$ versus Π_1. The full nonlinear term of the palinstrophy balance (Theorem A2) is $\Pi_1 = 2\mathcal{P}_1 + \int (H\omega_x)\omega\omega_x$. The production-balance analysis of this section applies the generator to $\mathcal{P}_1$ *itself*: $\mathcal{P}_1$ is the functional we differentiate (the production observable), whereas Π_1 is a term *in* the balance *of* the palinstrophy $\mathcal{E}_1$ — two distinct roles. We choose $\mathcal{P}_1$ rather than Π_1 because $2\mathcal{P}_1$ is the part of Π_1 whose own balance produces the sign-definite term W_1 below; applying the generator to the secondary term $\int (H\omega_x)\omega\omega_x$ yields no sign-definite leading term. This choice is part of the grouping-dependence disclosed in §5 (Normalization and interpretation).

The functional and its inertial balance (proved). $\mathcal{P}_1 = \int (H\omega)\omega_x^2$, $D\mathcal{P}_1 = -H(\omega_x^2) - 2(H\omega_x)\omega_x - 2(H\omega)\omega_{xx}$ (Lemma C.2; certified against finite differences, Appendix A). The inertial part $I(a) = \langle b_{nl}, D\mathcal{P}_1\rangle$, $b_{nl} = -au\omega_x + v\omega$, is (Proposition C.3, proved for general fields in Appendix C; certified to machine precision, Appendix A; $v = H\omega$, $v_x = H\omega_x$, $u = -\Lambda^{-1}\omega$):

$$I(a) = (2-2a)\int v^2\omega_x^2 - 2a\int uv\,\omega_x\omega_{xx} + 2\int vv_x\,\omega\omega_x + a\int u\,\omega_x\,H(\omega_x^2) - \int v\,\omega\,H(\omega_x^2). \tag{8}$$

At $a = -2$ the leading term is the **sign-definite** production-of-production $W_1 := 6\int (H\omega)^2\omega_x^2 \geq 0$; the remainder $-N_1 := I - W_1$ is nonlocal (carries u and H; displayed explicitly in Appendix C).

The projected balance (exact at every truncation). On E_M the generator pairs the drift with the *projected* gradient $P_M D\mathcal{P}_1$ — and, unlike the test functions of the §3 balances, $D\mathcal{P}_1$ is quadratic and leaves E_M. The exact nonlinear contribution of the Galerkin generator is therefore the projected pairing

$$I_M := \langle P_M N[\omega],\, P_M D\mathcal{P}_1\rangle \;=\; I - R_M, \qquad R_M := \langle Q_M N[\omega],\, Q_M D\mathcal{P}_1\rangle, \quad Q_M = \mathrm{Id} - P_M, \tag{9}$$

and stationarity with Injection Lemma (injection $\equiv 0$) gives, **exactly at every fixed truncation**,

$$E_\mu[I_M] = -\mathcal{C}_1, \qquad \text{equivalently} \qquad E_\mu[N_1] = E_\mu[W_1] + \mathcal{C}_1 - E_\mu[R_M], \tag{10}$$

with $\mathcal{C}_1 := \nu\, E_\mu[\langle \omega_{xx}, D\mathcal{P}_1\rangle]$ (no projection is needed in the viscous pairing: $\omega_{xx} \in E_M$), where $\mathcal{C}_1$ is the **viscous contribution** (measured negative at the small- and intermediate-ν operating points; at large ν the direct estimator is SNR-limited, Appendix A, and the sign is supplied by Theorem F for $M \geq 2k_f$; we call it the viscous *destruction* where hypothesis (NEG) of §5 is in force). The defect R_M is structurally small:

Lemma 4.1 (Support lemma for the Galerkin defect)**.** *If ω is supported on modes $|k| \leq K$ with $2K \leq M$, then $R_M(\omega) = 0$. In particular, $E_G[R_M] = 0$ for the Ornstein–Uhlenbeck Gaussian of a forcing band $k_f \leq M/2$.*

Proof. $N[\omega]$ and $D\mathcal{P}_1(\omega)$ are quadratic in ω, hence supported on modes $|k| \leq 2K \leq M$; Q_M annihilates both factors. □

Remark 4.2 (The defect is not identically zero: $M = 1$). For $\omega = A\cos x$ and $M = k_f = 1$: $N[\omega] = \frac{3}{2}A^2\sin 2x$ and $D\mathcal{P}_1 = \frac{5}{2}A^2\sin 2x$ are pure second harmonics, so $P_1 N[\omega] = 0$ — the one-mode Galerkin system is a *linear* OU system with $I_M \equiv 0$ and $\mathcal{C}_1 = 0$ — while $I = \frac{15\pi}{4}A^4$ and $W_1 = \frac{9\pi}{2}A^4$, so $I/W_1 = \frac{5}{6} \neq 0$. The full pairing I is therefore *not* the generator contribution in general, and the two normalized objects below genuinely differ at small M. (Certified in `anc/certify_projected_generator.py`, Appendix A.)

Two normalized objects. We accordingly distinguish

$$\theta := 1 - \frac{E_\mu[N_1]}{E_\mu[W_1]} = \frac{E_\mu[I]}{E_\mu[W_1]} \qquad \text{and} \qquad \theta_M := \frac{E_\mu[I_M]}{E_\mu[W_1]} = -\frac{\mathcal{C}_1}{E_\mu[W_1]}, \tag{11}$$

the **normalized production diagnostic** (grouping-dependent, scale-blind — Lemma 6.1 — and directly measurable: the object of all measured curves and of Theorem F's first bound) and the **exact balance coefficient** of the truncated system. By (10) they differ by the Galerkin defect, $\theta - \theta_M = E_\mu[R_M]/E_\mu[W_1]$, which we measure at every operating point (Table 1): from $\sim 10^{-8}$ at $\nu \geq 0.05$ through $\approx 3 \cdot 10^{-5}$ at $\nu = 0.01$ to $\approx 5 \cdot 10^{-3}$ at $\nu = 0.005$, where the spectrum fills toward the cutoff and R_M becomes a genuine truncation boundary term. The two objects respond differently to the cutoff: at $\nu = 0.005$ the defect drops from $4.7 \cdot 10^{-3}$ ($M = 85$, pooled) to $5.3 \cdot 10^{-5}$ ($M = 127$) — two orders of magnitude for a 1.5× cutoff increase — while θ itself is unchanged (0.7722 at both cutoffs, within one band): θ is M-stable, while θ_M is the balance coefficient *of the particular truncated system*: in the two cutoff checks performed, increasing M reduced the positive normalized defect while θ stayed within one statistical band (we do not claim a proved sign or monotonicity of $E_\mu[R_M]$ in M). This is why we report θ as the primary measured object and θ_M alongside it. A genuine distinction of objects, not a convention.

Measured (proved-supported) — the diagnostic is positive and ν-stable. On the stationary state sampled by the solver at $a = -2$ (Appendix A; 2 seeds each; the campaign of Table 1 and Figure 2):

$$\theta = 0.789 \ (\nu = 0.05), \quad 0.781 \ (\nu = 0.02), \quad 0.776 \ (\nu = 0.01). \tag{12}$$

$\theta \in (0, 1)$ — two separate observations: $\mathcal{C}_1 < 0$ is measured, which by the exact balance (10) gives $\theta_M > 0$; and $E_\mu[I] > 0$, $E_\mu[N_1] > 0$ are measured, giving $0 < \theta < 1$ directly — and θ **decreases only gently across a decade of ν, toward a finite positive limit** — empirical support for a floor $\theta \geq \theta_0 > 0$ (§6). *(The primary dataset for Table 1 and Figure 2 is the exact-OU campaign* `anc/p2_results_v18.jsonl`*, and the three values above are its pooled per-ν means. These are 2-seed points; the within-run $n = 2$ standard error is uninformative and is not quoted — the per-seed jackknife band of Table 1 is* ± 0.0006*, and the cross-code spread reported next is a separate, systematic scatter.)* An **independent re-measurement on separately written code** (a plain Euler–Maruyama solver, blind to the integrating-factor implementation; Appendix A) gives $\theta = 0.807, 0.771, 0.781, 0.775$ at $\nu = 0.05, 0.02, 0.01, 0.005$, agreeing with the campaign to $\lesssim 2.3\%$ (largest at $\nu = 0.05$, where the two time-steppers' $O(dt)$ biases differ most), and shows θ **varying mildly within** $[0.77, 0.81]$ **across the small-ν range** $\nu \in [0.005, 0.2]$ (Figure 2) (the $\nu = 0.01$ point fully resolved, dealiased tail $3 \cdot 10^{-11} \times$ peak); above $\nu \approx 0.5$ it climbs to the Gaussian anchor $\theta_G \approx 0.812$ (§4, large-ν). *(θ computed w.r.t. the leading square $W_1 = 6 \int v^2 \omega_x^2$; an alternative admissible grouping $W_1' = c\,W_1$ rescales the value ($\theta' = \theta/c$, which for $c < 1$ may exceed 1) but preserves the qualitative sign conclusion $\theta > 0$ and its ν-stability. The two-sided bracket $\theta \in (0, 1)$ is reported for the stated W_1-grouping.)*

Stationarity-controlled campaign. Long exact-OU-stepped runs (burn-in $\geq 5/\nu$, stationary windows $\geq 20/\nu$, two seeds per point, block statistics; split-half drift z reported *on the observables* *themselves* — $z := (\bar{X}_1 - \bar{X}_2)/\mathrm{SE}(\bar{X}_1 - \bar{X}_2)$, the standardized difference of the first- and second-half block means of an observable X over the jackknife standard error of that difference, so $|z| < 2$ detects no statistically resolvable drift (a diagnostic, not a proof of stationarity) — per-run controls: L_0 ratio $\in [0.95, 1.06]$, dealiased tail $\leq 9 \cdot 10^{-7}\times$ peak, hereditary-lemma $\Pi_{LL} \sim 10^{-19}$; parameters in Table 3, Appendix A):

Table 1: The stationary campaign (strict-mask solver, padded diagnostics; two seeds per point; pooled θ = ratio of the seeds' combined sample means, uncertainty = per-seed block jackknife). $\theta - \theta_M = E[R_M]/E[W_1]$ = the normalized Galerkin defect (§4); $\delta_{\mathrm{bal}} = |E[I_M] + \mathcal{C}_1|/E[W_1]$ = the balance residual (dominated by the statistical error of the noisy $\mathcal{C}_1$ estimator); split-half $\max|z|$ over $I, W_1, \mathcal{E}_1, D$. Padded-vs-unpadded aliasing receipts: $|\Delta\theta| \leq 4 \cdot 10^{-6}$ at every point. (R_M is measured positive at every operating point, so the signed column $\theta - \theta_M$ and the absolute defect δ_M of Appendix A coincide.) Data: `anc/p2_results_v18.jsonl`.

ν	θ (pooled)	$\theta - \theta_M$	δ_{bal}	$\max\|z\|$	$\rho_\nu(\{\|k\| < 5\})/(\frac{1}{2}B_0)$	HH share of $\Pi(5)$
0.2	0.8061 ± 0.0005	$< 10^{-12}$	$1.7 \cdot 10^{-2}$	1.6	0.997	0.005
0.1	0.7978 ± 0.0006	$< 10^{-12}$	$9 \cdot 10^{-3}$	1.5	0.941	0.023
0.05	0.7890 ± 0.0006	$2.3 \cdot 10^{-8}$	$3.1 \cdot 10^{-3}$	1.2	0.763	0.064
0.02	0.7808 ± 0.0004	$1.4 \cdot 10^{-6}$	$1.7 \cdot 10^{-3}$	1.7	0.424	0.139
0.01	0.7758 ± 0.0003	$3.3 \cdot 10^{-5}$	$2.4 \cdot 10^{-3}$	0.6	0.243	0.194
0.005	0.7722 ± 0.0001	$4.7 \cdot 10^{-3}$	$2.5 \cdot 10^{-3}$	0.9	0.128	0.231

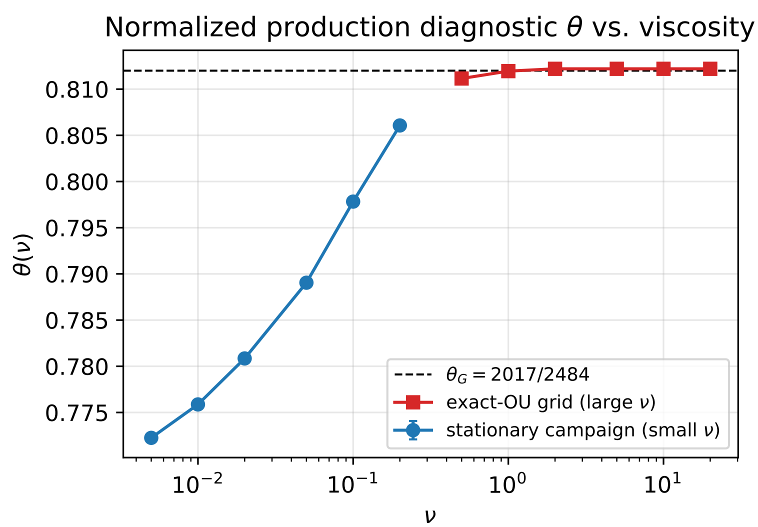
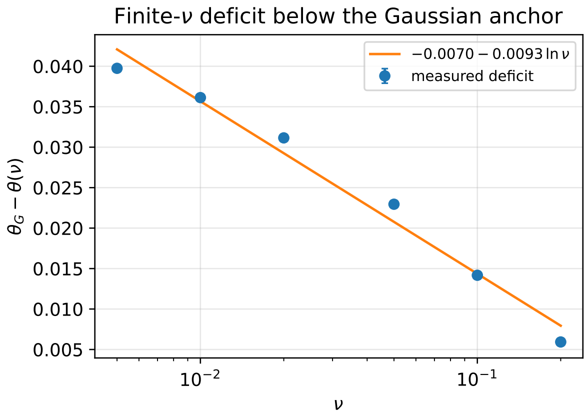


Figure 2: The production-balance diagnostic $\theta(\nu)$: a small finite-ν deficit below the Gaussian anchor $\theta_G = 2017/2484$ (stationary campaign, small ν), climbing to θ_G by $\nu \approx 0.5$ (exact-OU grid, large ν). Two seeds per point. Data: `anc/p2_results_v18.jsonl`, `anc/theta_large_nu_exact_ou.log`.

($dt/2$ check at $\nu = 0.01$: $\theta = 0.7754 \pm 0.0003$, consistent; the full step/resolution convergence matrix is reported in Appendix A, Table 4.) Three findings. (i) The finite-ν deficit below the exact Gaussian anchor $\theta_G = 2017/2484$ is resolved and is well described by a **logarithm** over the measured range (Figure 3): $\theta_G - \theta(\nu) \approx -0.0070 - 0.0093 \ln\nu$ (coefficient s.e. 0.0025, 0.0007; max residual 0.0025, at the endpoint $\nu = 0.005$) — a purely descriptive fit over the six points $\nu \in [0.005, 0.2]$, asserting no functional form and claiming no extrapolation. (ii) The low-mode dissipation share drains with an *increasing* local rate: $\rho_\nu \sim \nu^{\alpha_{\mathrm{loc}}}$ with $\alpha_{\mathrm{loc}} = 0.08 \to 0.30 \to 0.64 \to$



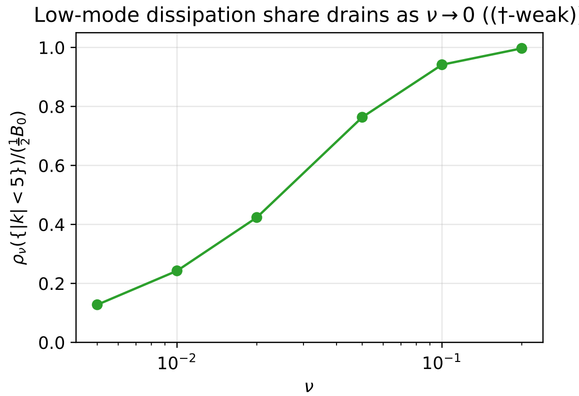


Figure 3: Left: the finite-ν deficit $\theta_G - \theta(\nu)$ is well described by a logarithm $\approx -0.0070 - 0.0093 \ln \nu$ over $\nu \in [0.005, 0.2]$ (descriptive fit, no extrapolation claimed). Right: the low-mode dissipation share $\rho_\nu(\{|k| < 5\})/(\frac{1}{2}B_0)$ drains from ≈ 1 to 0.128 as $\nu \to 0$ — the empirical form of (†-weak) with a rate. Data: `anc/p2_results_v18.jsonl`.

$0.80 \to 0.93$ between adjacent ν (propagated s.e. ≤ 0.03 per exponent). These finite-difference exponents are sensitive to statistical and resolution error and are reported as an empirical trend consistent with (†-weak), not as evidence of a limit $\alpha \to 1$. (iii) The negative inertial-generation fraction is 0 on every stationary sample (n up to 8000 per run): $I > 0$ pointwise along every recorded trajectory.

The Gaussian anchor: θ_G is the large-ν limit. Expand the Wick (Gaussian) expectations $E_G[I]$ and $E_G[W_1]$ over the rescaled OU stationary shape in the per-mode variances s_k^2: by independence only the diagonal (s_k^4) and cross ($s_i^2 s_j^2$) monomials survive, with coefficients $c_{\bullet,kk}$ and $c_{\bullet,ij}$ (the explicit Wick sums and the degenerate Gaussian $G_\nu = \mathcal{N}(0, \mathrm{diag}(b_k^2/(2\nu k^2)))$ they arise from are written out in Lemma B.6, Appendix B). The **cross-mode coefficient ratio**

$$\theta_G(m) := c_{I,ij}/c_{W,ij} = (5 + 7m + 3m^2)/(3 + 12m + 3m^2), \qquad m = j/i \ (i < j), \tag{13}$$

is the pairwise-interaction ratio of a single mode pair — *not* the ratio $E_G[I]/E_G[W_1]$ of a two-mode field (certified, Appendix A). It has global minimum $\theta_{G,\min} = 1/3 + \sqrt{69}/18 \approx 0.7948$ at the non-integer $m^* = (2 + \sqrt{69})/5 \approx 2.06$ (minimality independently re-verified, Appendix A). (As a bare rational function $\theta_G(m)$ is *not* bounded by 1 — it reaches 5/3 as $m \to 0$ and crosses 1 at $m = 2/5$; the $\theta_G \in [\theta_{G,\min}, 1)$ range relies on the physical convention $m = j/i > 1$ $(i < j)$, where $m^* = 2.06$ is the interior minimizer.) The full-band value $\theta_G(\text{band}) = E_G[I]/E_G[W_1]$ is the c_W-weighted mediant of the diagonal self-ratio $c_{I,kk}/c_{W,kk} = 5/6$ and the cross ratios $\theta_G(j/i)$; since every weight $c_{W,\bullet} > 0$ and every ratio is $\geq \theta_{G,\min}$ (verified for band $[1,4]$: the minimum attained over its integer pairs is $\theta_G(2) = 31/39 = 0.7949$ at $m = 2$, just above the family infimum 0.7948 at the unattainable $m^* \approx 2.06$; diagonal 5/6), the band value provably exceeds the floor. For the *flat* (equal-amplitude) forcing band $[1,4]$ the Gaussian value is the exact rational number $\theta_G(\text{flat band}) = 2017/2484 = 0.811996779\ldots$ (exact Wick sums in rational arithmetic; confirmed by direct sampling of the Ornstein–Uhlenbeck stationary law to $5.6 \cdot 10^{-5}$; scripts in Appendix A). Because θ is a ratio of two quartic functionals (hence 0-homogeneous in the field amplitude) and the rescaled OU stationary shape $s_k^2 \propto b_k^2/k^2$ is ν-independent, $\theta_G(\text{band})$ is the $\nu \to \infty$ **limit of** θ (proved at the Galerkin level, Theorem F §6; the solver evolves such a system exactly in space, with the nonlinear drift advanced explicitly in time), and this is what the solver measures: with exact-OU noise stepping, $\theta = 0.8111$–0.8123 (CI95 half-widths $\lesssim 4 \cdot 10^{-4}$, split-half z clean, $dt/5$-stable) *uniformly* across $\nu \in \{0.5, 1, 2, 5, 10, 20\}$, two seeds each. The small-ν points (0.772–0.789 over $\nu \in [0.005, 0.05]$) thus read as a **small finite-ν non-Gaussian deficit below the Gaussian anchor**, from which $\theta(\nu)$ climbs back to θ_G by $\nu \approx 0.5$. The small-ν deficit is resolved by the stationarity-controlled campaign below, reaching ≈ 0.040 at $\nu = 0.005$.

Injection Lemma (Homogeneous-noise injection vanishes)**.** *For homogeneous noise $b_k = f(|k|)$, the Itô injection into the palinstrophy production balance vanishes:* $\frac{1}{2}\mathrm{Tr}(QD^2\mathcal{P}_1) = 0$.

Proof. $D^2\mathcal{P}_1(h,h) = 2\int(H\omega)h_x^2 + 4\int(Hh)\omega_x h_x$. The individual $\cos_k$ and $\sin_k$ trace summands do *not* separately vanish; rather, at the common homogeneous amplitude b_k the two members of each $(\cos_k, \sin_k)$ pair enter the trace with equal and opposite weight, so each $|k|$-pair cancels and the total trace is 0. Homogeneity is essential: with unequal amplitudes on the pair the trace is genuinely nonzero (certified below). □

(Certified: homogeneous 10^{-16}, anisotropic $\neq 0$; Appendix A.) Thus the production balance (10) is injection-free: $E_\mu[N_1] = E_\mu[W_1] + \mathcal{C}_1 - E_\mu[R_M]$, with no Itô term.

5 Reduction, conditional bound, density (corollaries of the production balance)

Given the certified production balance of §4, the following consequences are proved below; all proofs are self-contained and rest only on the standing hypotheses (PB), (NEG), (POS) stated next (each result names which of them it uses). These are corollaries of the balance — an algebraic reduction (Corollary C) and a moment inequality (Corollary E) — recorded to delimit what the diagnostic θ does and does not control; the paper's main new theorem is the large-viscosity limit (Theorem F, §6).

Standing hypotheses

- **(PB)** Production balance (exact at every truncation via the projected pairing, eq. (10)): for every invariant μ of the Galerkin system, $E_\mu[I_M] = -\mathcal{C}_1$, i.e. $E_\mu[N_1] = E_\mu[W_1] + \mathcal{C}_1 - E_\mu[R_M]$, where $W_1 \geq 0$ pointwise is the production-of-production, N_1 the nonlocal coherence term, and R_M the measured Galerkin defect (Lemma 4.1; per-point values in Table 1, from $\sim 10^{-8}$ to $\approx 5 \cdot 10^{-3}$ at the smallest ν). The noise injection is **absent** (Injection Lemma: homogeneous-noise Itô term $\equiv 0$, certified).

- **(NEG)** $\mathcal{C}_1 \leq 0$ (the mean viscous destruction is nonpositive). *Status: a hypothesis, not a theorem* — §6 shows the sign of $\mathcal{C}_1 = \nu E_\mu[Y_\mathrm{d}]$ with $Y_\mathrm{d} := \langle \omega_{xx}, D\mathcal{P}_1 \rangle$ exactly odd under $\omega \to -\omega$ is a property of the measure's asymmetry, not a pointwise inequality. It is *measured* to hold cleanly at the small- and intermediate-ν operating points (§4); at large ν the direct estimator is SNR-limited and the sign is *not* statistically resolved there ($\nu = 10$: $\mathcal{C}_1 = -2.1 \cdot 10^{-4} \pm 2.7 \cdot 10^{-4}$) — positivity of θ_M at large ν is a *theorem* at the Galerkin level for $M \geq 2k_f$, with a strictly positive margin ($\theta_M \to \theta_G > 0$, Theorem F, §6), though the theorem does not supply an explicit threshold ν_1. Under (NEG), $E_\mu[N_1] \leq E_\mu[W_1]$.

- **(POS)** $E_\mu[W_1] > 0$ for every invariant measure. *At a fixed Galerkin truncation this is a theorem, not a hypothesis* (Lemma 5.1).

Lemma 5.1 (Positivity of the production denominator)**.** *At a fixed Galerkin truncation, $E_\mu[W_1] > 0$ for every invariant measure μ.*

Proof. $W_1 = 6\int (H\omega)^2\omega_x^2 = 6\|(H\omega)\,\omega_x\|^2 \geq 0$, so $W_1 = 0$ forces $(H\omega)\,\omega_x \equiv 0$. Since $H\omega$ and ω_x are real-analytic trigonometric polynomials, the zero set of each is finite unless it vanishes identically; hence $H\omega \equiv 0$ or $\omega_x \equiv 0$, and either gives $\omega \equiv$ const, so $\omega \equiv 0$ by zero mean. Thus $W_1 = 0$ only at $\omega = 0$, and $E_\mu[W_1] = 0$ would force $\mu = \delta_0$; but δ_0 is not invariant under the nonzero additive noise ($B_0 > 0$). Hence $E_\mu[W_1] > 0$. $\square$

Definition 5.2 (Production diagnostic and balance coefficient)**.**

$$\theta := 1 - \frac{E_\mu[N_1]}{E_\mu[W_1]} = \frac{E_\mu[I]}{E_\mu[W_1]}, \qquad \theta_M := -\frac{\mathcal{C}_1}{E_\mu[W_1]} = \theta - \frac{E_\mu[R_M]}{E_\mu[W_1]} \quad (\S4), \tag{14}$$

well-defined real ratios under (POS); $\theta_M \geq 0$ iff (NEG) holds, and $\theta \leq 1$ iff $E_\mu[N_1] \geq 0$ (a measured, not proved, sign). $\theta_M = 0$ exactly when the destruction vanishes (steady/laminar trap); measured $\theta \in (0,1)$ at every operating point with $\theta - \theta_M$ per point in Table 1 ($\leq 0.7\%$ of θ everywhere), and $\theta \to \theta_G(shape) \approx 0.812$ as $\nu \to \infty$ (§4).

Normalization and interpretation. The ratio θ is defined *relative to the specific production grouping* $W_1 = 6\int (H\omega)^2\omega_x^2$ (the full, sign-definite $\int v^2\omega_x^2$ term of I at $a = -2$). This grouping is natural but not proved canonical: a different admissible split $I = W_1' - N_1'$ rescales the denominator and hence the numerical value of θ; only the *unnormalized* means $E_\mu[I]$ and $E_\mu[I_M] = -\mathcal{C}_1$ are grouping-invariant. We therefore read θ as a **normalized, grouping-dependent production-balance diagnostic**, not a canonical physical efficiency; the name "cascade efficiency" of earlier versions is retired in favor of the neutral name. In particular (§6, paragraph (S6)) a positive $\theta = O(1)$ is compatible with the *vanishing absolute-production limit* $E_\mu[W_1], |\mathcal{C}_1| \to 0$: θ measures the *relative* balance of production against coherence, not the magnitude of an absolute cascade. A canonical grouping (or a variational derivation of W_1) is open.

Corollary C (Reduction)**.** *Fix $\varepsilon \in (0,1)$. For any invariant μ (Galerkin-rigorous; formal for (SG), §2; assuming (POS), so that the θ-forms are well-defined),*

$$E_\mu[N_1] \leq (1-\varepsilon)E_\mu[W_1] \quad \iff \quad \theta \geq \varepsilon \quad \iff \quad -\mathcal{C}_1 + E_\mu[R_M] \geq \varepsilon E_\mu[W_1], \tag{15}$$

the last equivalence via the exact balance (10)*; in terms of the balance coefficient, $\theta_M \geq \varepsilon \iff -\mathcal{C}_1 \geq \varepsilon E_\mu[W_1]$. A mean-coherence bound of margin ε is EXACTLY a floor $\theta \geq \varepsilon$ on the diagnostic; up to the measured defect $E_\mu[R_M]$, a lower bound of $\varepsilon E_\mu[W_1]$ on the viscous destruction $-\mathcal{C}_1$.*

Proof. The first equivalence is algebra ($I = W_1 - N_1$ pointwise, (POS)): $E_\mu[N_1] \leq (1-\varepsilon)E_\mu[W_1] \iff E_\mu[I] \geq \varepsilon E_\mu[W_1] \iff \theta \geq \varepsilon$. The second substitutes $E_\mu[I] = -\mathcal{C}_1 + E_\mu[R_M]$ from (10). $\square$

Content. $-\mathcal{C}_1$ is the *viscous* destruction of the palinstrophy production — a purely dynamical object (it vanishes on the steady trap) — and $E_\mu[R_M]$ is the measured Galerkin defect ($\leq 0.7\%$ relative at our operating points, Table 1). No kinematic/pointwise inequality can produce the margin: it lives in the cascade dynamics. This localizes precisely what a proof of $\theta \geq \theta_0$ must supply.

Remark 5.3 (Collapse of the conditional coherence bound). In a multi-component setting one would typically compare the coherence numerator N against a *separate* dissipation-comparable object D ($W \geq D$, $\beta := E[D]/E[W] \in (0,1]$), giving $R = E[N]/E[D] = (1-\theta)/\beta$ and a margin

$\varepsilon = 1 - (1-\theta_0)/\beta_{\min}$ under two floors $\theta \geq \theta_0$, $\beta \geq \beta_{\min}$. In the 1D palinstrophy production balance *as constructed here* there is no independent dissipation-comparable at the same order: the production-of-production W_1 and the coherence N_1 are the only two $O(1)$ objects in the balance (Def θ), the destruction is $\mathcal{C}_1 = -E_\mu[I]$ (viscous), and no such D_1 arises without introducing the palinstrophy-dissipation $\nu \int \omega_{xx}^2$, which lives at a different order (the Level-1 balance, Theorem A2 of §3). (This is an observation about the term inventory of the present construction, not an impossibility theorem over a defined class.) Consequently the conditional coherence bound is **exactly Corollary C**: a floor $\theta \geq \theta_0$ gives, with no β apparatus,

$$E_\mu[N_1] \leq (1-\theta_0)\, E_\mu[W_1]. \tag{16}$$

Measured (§4, stationary campaign): $\theta \in [0.776, 0.789]$ across $\nu \in [0.01, 0.05]$, so $E_\mu[N_1] \leq 0.224 \cdot E_\mu[W_1]$ at the measured operating points — the coherence is comfortably sub-production. The 1D balance is *cleaner* here (one fewer hypothesis: no $\beta_{\min}$). **The genuinely open input is the $\theta_0 > 0$ floor**, which §6 shows requires a *scale-resolved* efficiency, not this total-magnitude one.

Corollary E (Density of negative inertial-generation episodes, moment form)**.** *Define the **negative inertial-generation set** $\mathcal{A} := \{W_1 \leq N_1\}$ — states where the nonlocal coherence overwhelms the local production-of-production (θ is locally negative). Let $Y := W_1 - N_1 = I$ (pointwise), so $E_\mu[Y] = E_\mu[I] = \theta \cdot E_\mu[W_1]$ ($= -\mathcal{C}_1 + E_\mu[R_M]$ by (10)). Suppose (POS), $\theta > 0$ (measured at every operating point; a theorem at large ν for $M \geq 2k_f$, Theorem F), and $E_\mu[Y^2] < \infty$ (automatic at any fixed Galerkin truncation, Lemma B.2). Then*

$$\mu(\mathcal{A}) \;\leq\; 1 - \frac{(E_\mu[Y])^2}{E_\mu[Y^2]} \;=\; 1 - \frac{\theta^2\, E_\mu[W_1]^2}{E_\mu[Y^2]} \;<\; 1, \tag{17}$$

and, if μ is ergodic*, by Birkhoff's ergodic theorem the time-density of negative inertial-generation episodes along μ-typical trajectories obeys the same bound. For a non-ergodic invariant μ the time-average along a typical trajectory equals the conditional expectation $E_\mu[\mathbf{1}_{\mathcal{A}} \mid \mathcal{I}]$, so the bound holds trajectory-wise only component-by-component (with each ergodic component's own θ, $E[W_1]$, $E[Y^2]$); the displayed bound then controls the μ-average of these densities. (At small ν, where uniqueness/ergodicity is open, this distinction is substantive and we do not elide it.)*

Proof. Under $\theta > 0$, $E_\mu[Y] = \theta E_\mu[W_1] > 0$. By Cauchy–Schwarz, $E_\mu[Y] \leq E_\mu[Y \mathbf{1}_{Y>0}] \leq \sqrt{E_\mu[Y^2]\, \mu(Y>0)}$, hence $\mu(Y>0) \geq (E_\mu[Y])^2/E_\mu[Y^2]$ and $\mu(\mathcal{A}) = \mu(Y \leq 0) = 1 - \mu(Y>0) \leq 1 - (E_\mu[Y])^2/E_\mu[Y^2]$. Birkhoff applies since $\mathbf{1}_{\mathcal{A}}$ is bounded measurable. $\square$

Remark 5.4 (L^∞ form). If additionally the excursion bound $B := \operatorname{ess\,sup}_\mu Y < \infty$ holds, with $b := B/E_\mu[W_1]$, then $\mu(\mathcal{A}) \leq 1 - \theta/b < 1$: for any r.v. with $Y \leq B$, $E[Y] = E[Y; Y \leq 0] + E[Y; Y > 0] \leq B\, \mu(Y>0)$, so $\mu(Y \leq 0) \leq 1 - \theta/b$ (non-vacuity $\theta \leq b$ is automatic). However, with additive Gaussian forcing the invariant measure is expected to have unbounded support on the forced modes, so B is plausibly **infinite** and is in any case not accessible to finite simulation (a sample maximum estimates a quantile, never an essential supremum): the L^∞ form is a structural statement only, and Corollary E in its moment form is the quantitative deliverable.

Reading. A positive diagnostic θ forces the negative inertial-generation set to have sub-unit measure. The inputs are $\theta > 0$ (§6) and a second moment of Y, measurable at the operating points. Measured (the §4 stationary campaign, exact-OU-stepped, two seeds pooled per point; note $Y = W_1 - N_1 = I$ pointwise): the bound gives $\mu(\mathcal{A}) \leq 0.78$ at $\nu = 0.05$ and ≤ 0.80 at $\nu = 0.01$ — valid but far from tight: the *empirical* negative inertial-generation fraction is 0 on every one of the 8k pooled stationary samples per point ($I > 0$ pointwise along the recorded trajectories; the samples are time-correlated, so we report the observed fraction only and do not convert it into a bound on $\mu(\mathcal{A})$). Given this large gap between the loose bound (≤ 0.78) and the observed fraction (0), Corollary E is best read as a *qualitative* structural guarantee (a positive diagnostic forces the negative inertial-generation set below full measure) rather than a sharp quantitative estimate. The observed fraction 0 on a finite, time-correlated sample is *not* evidence that $\{I \leq 0\}$ has zero measure, and does not establish kinematic positivity of I; whether $I > 0$ holds kinematically on band-limited fields is open, and would require a targeted search (e.g. minimizing I/W_1 over band-limited fields) to decide. (The quoted bounds are computed from the §4 stationary campaign's padded diagnostics, $E[Y^2]$ recorded per point.)

What is proved vs pending

statement	status
Def θ, θ_M (real ratios, under (POS)); (PB) exact via the projected pairing (10)	**proved** (Injection Lemma certified; Lemma 4.1 gives an exact support criterion for R_M's vanishing and Lemma B.5 the large-ν bound; $\theta - \theta_M$ measured $\leq 0.7\%$ of θ, $\approx 5 \cdot 10^{-3}$ at the smallest ν); canonical W_1/N_1 grouping open
$\theta \in [0, 1]$, i.e. (NEG) + $E_\mu[N_1] \geq 0$	**hypotheses**: measured at all operating points; (NEG) a theorem at large ν, $M \geq 2k_f$ ($\theta_M \to \theta_G > 0$, Thm F)
Cor C (reduction)	**proved** (algebra + (10))
Remark 5.3 (conditional bound)	collapses to Cor C (no independent D_1 at this order in the present construction)
Cor E (negative inertial-generation density, moment form)	**proved** (given (POS), $\theta > 0$; second moment automatic at Galerkin; ergodicity needed for the trajectory-wise reading)
the excursion bound $b < \infty$, uniform in ν	plausibly false ($B = \infty$); the L^∞ form is Remark 5.4 only (measured moment bounds 0.78/0.80; empirical negative inertial-generation fraction 0)
$\theta \geq \theta_0 > 0$ uniformly in ν	OPEN at small ν (§6); at large ν **proved at Galerkin level,** $M \geq 2k_f$ (Thm F: $\theta, \theta_M \to \theta_G = 2017/2484$, so (NEG) is a theorem for $\nu \geq \nu_1$)

6 The diagnostic floor: obstructions and the large-viscosity theorem

We attacked $\theta \geq \theta_0 > 0$ and the flux-law input by four independent strategies (verification record in Appendix A). The honest outcome is **one new unconditional theorem, a strict weakening of the open input, a certified structural lemma, and a scale-blindness lemma** delimiting what the scalar θ can deliver — not an unconditional $\theta_0 > 0$, which meets a genuine turbulence-closure obstruction.

(S1) Unconditional — flux law as UV-concentration (Corollary A1″, §3). The two-hypothesis chain collapses to a hypothesis-free equivalence: (FL∞) $\iff$ ρ_ν UV-concentrates.

This *replaces* the earlier over-hypothesized corollary.

(S2) Weakening — (†) → (†-weak). The open input is $E_\mu\|P_{<K}\omega\|^2 = o(1/\nu)$, not $O(1)$; the naive Poincaré bound sits exactly at the borderline exponent, so only an $o(1)$ gain is needed.

(S3) Certified structural lemma — hereditary advective cancellation. The $a = -2$ cancellation is not a full-field accident: the low-mode Galerkin self-interaction vanishes for *every* truncation, $\langle \omega_L, P_{<K}N[\omega_L]\rangle \equiv 0\ \forall K$ (certified 24/24 to $\leq 9 \cdot 10^{-14}$, Appendix A). *(It is a one-line corollary of the A0 cancellation applied to ω_L via the self-adjoint idempotent $P_{<K}$; the value is the decomposition it enables, not the proof depth.)* Hence the flux $\Pi(K)$ is carried entirely by cross/high-mode leakage (a linear-in-$\|\omega_H\|$ term plus a quadratic one), which localizes what any proof of (†-weak) must control.

(S4) The scale-blindness obstruction — why θ is not the bridge. The obstruction is now a lemma. For $\lambda \in \mathbb{N}$ let S_λ denote the *mode dilation* $(S_\lambda\omega)(x) = \omega(\lambda x)$, i.e. $\widehat{S_\lambda\omega}(\lambda k) = \hat\omega(k)$ (and $= 0$ off $\lambda\mathbb{Z}$).

Lemma 6.1 (Scale-blindness of θ)**.** *S_λ commutes with H and with pointwise products, and satisfies $\partial_x S_\lambda = \lambda\, S_\lambda \partial_x$ and $\Lambda^{-1}S_\lambda = \lambda^{-1}S_\lambda\Lambda^{-1}$. Consequently every monomial of I and of W_1 is homogeneous of total dilation weight 2, so for every field*

$$I(S_\lambda\omega) = \lambda^2\, I(\omega), \qquad W_1(S_\lambda\omega) = \lambda^2\, W_1(\omega). \tag{18}$$

Hence for every probability measure μ on field space (finite fourth moments, $E_\mu[W_1] > 0$) and its dilation pushforward $\mu_\lambda := (S_\lambda)_*\mu$:

$$\theta(\mu_\lambda) = \frac{E_{\mu_\lambda}[I]}{E_{\mu_\lambda}[W_1]} = \frac{\lambda^2 E_\mu[I]}{\lambda^2 E_\mu[W_1]} = \theta(\mu) \qquad \textit{for every } \lambda, \tag{19}$$

while the low-mode dissipation share is not *dilation-invariant: if μ is supported on modes $|k| \leq K_0 < K$, the share $\rho_{\mu_\lambda}([-K,K))/\rho_{\mu_\lambda}(\mathbb{Z}^*)$ equals 1 at $\lambda = 1$ and 0 for every $\lambda > K$.*

Proof. On Fourier coefficients S_λ relabels $k \mapsto \lambda k$: the H multiplier $-i\,\mathrm{sgn}(\lambda k) = -i\,\mathrm{sgn}(k)$ ($\lambda > 0$) commutes with the relabeling (weight 0); each ∂_x contributes one factor λ (weight 1); Λ^{-1} contributes $1/|\lambda k| = \lambda^{-1}/|k|$ (weight -1); products are pointwise, and $\int_0^{2\pi} f(\lambda x)\,dx = \int_0^{2\pi} f$ for $\lambda \in \mathbb{N}$. Assigning weights $\omega \mapsto 0$, $v = H\omega \mapsto 0$, $u = -\Lambda^{-1}\omega \mapsto -1$, each derivative $\mapsto +1$: every monomial of $I = (2-2a)\int v^2\omega_x^2 - 2a\int uv\,\omega_x\omega_{xx} + 2\int vv_x\omega\omega_x + a\int u\omega_x H(\omega_x^2) - \int v\omega H(\omega_x^2)$ and of $W_1 = 6\int v^2\omega_x^2$ has total weight 2 (0+0+1+1; −1+0+1+2; 0+1+0+1; −1+1+2; 0+0+2). The dissipation statement is immediate: ρ_{μ_λ} carries its mass on modes $\lambda|k| \geq \lambda$. □

Corollary 6.2 (No bound factoring through θ closes (†-weak))**.** *There is no function f with $\rho_\mu([-K,K))/\rho_\mu(\mathbb{Z}^*) \leq f(\theta(\mu)) < 1$ valid over all probability measures: the Ornstein–Uhlenbeck Gaussian G on the band $[1,2]$ has $\theta(G) \geq \theta_{G,\min} \approx 0.7948$ (the mediant argument of §4) and low-mode share* 1 *for $K > 2$, while its dilations G_λ have the* same *θ and share* 0 *for $\lambda > K$. Consequently any proof of* (†*-weak) from a floor $\theta \geq \theta_0$ must use the dynamical input that μ is an* invariant measure *of (SG) — information not contained in the value of θ. In this precise sense θ is scale-blind: bounds factoring through the scalar θ alone cannot control spectral localization.*

Scope. Lemma 6.1 and its corollary concern the **full diagnostic** $\theta = E[I]/E[W_1]$ (I, W_1 are functionals of the field alone, and the dilation argument never invokes a balance). The exact balance coefficient θ_M (§4) carries the projector P_M, which does *not* commute with S_λ; a dilation statement for θ_M would compare *different* truncated systems, $(\omega, M) \mapsto (S_\lambda\omega, \lambda M)$, and we do not claim it. Since the measured defect $\theta - \theta_M$ is $\leq 0.7\%$ of θ at every operating point (Table 1), the obstruction reading is unaffected.

Discussion (the mechanism behind Lemma 6.1). Three complementary routes anticipate the lemma (locality mismatch: $W_1 = 6\int v^2\omega_x^2$ is an all-scales, H-unimodular functional carrying no low-k weight; dimensional: a dimensionless θ cannot convert the ν^{-1}-divergent Poincaré bound into $o(\nu^{-1})$; derivational: no generator/Cauchy–Schwarz identity bridges the bilinear low-pass pairing to the cubic unfiltered ratio). The precise missing ingredient is a **scale-resolved efficiency** (a K-dependent floor on the rate at which $\rho_\nu(\{|k| < K\})$ is driven to $o(1/\nu)$) plus a dynamical (martingale/Grönwall) argument. This is a genuine turbulence-closure obstruction: the total-magnitude diagnostic is bounded and measured ($\theta \approx 0.78$, §4), but the *scale-resolved* residual that the flux law actually needs is not controlled by it. (Honest scope: Corollary 6.2 rules out bounds factoring through θ over the class of all measures; it does not exclude dynamical arguments that use invariance itself.)

A **further obstruction route** (independent of the above) operates at the level of θ's own integrand: the destruction integrand $Y_{\mathrm{d}} := \langle \omega_{xx}, D\mathcal{P}_1\rangle$ (an odd cubic — distinct from Corollary E's even quartic $Y = W_1 - N_1$) is **exactly odd under** $\omega \to -\omega$ (machine-zero check, Appendix A) and pointwise sign-indefinite (Appendix A) — so $-E_\mu[Y_{\mathrm{d}}] > 0$ (hence $\theta_M > 0$; the diagnostic θ then differs by the measured defect) is a property of the *measure's* $\omega \to -\omega$-asymmetry, not a kinematic inequality; no pointwise bound delivers it.

(S5) The measured evidence (§4). $\theta \approx 0.78$, ν-stable across the factor-40 range $\nu \in [0.005, 0.2]$ and $\theta \to \theta_G \approx 0.812$ at large ν — the diagnostic floor $\theta_0 > 0$ is empirically robust, but *proving* it remains open (§6, the paper's central open problem); closing (FL∞) additionally requires the *scale-resolved* input (†-weak), not merely a total-magnitude floor. A previously proposed positive mechanism — a phase-homogeneity hypothesis ($\star$): $E_\mu[\Pi_{HH}] = 0$, i.e. the mean high–high flux piece (§2) vanishes, which would remove the small-scale leakage into the low-mode dissipation and reduce (†-weak) by one order — is now **refuted as a small-ν mechanism** (pre-registered falsifier): in the stationary campaign the mean HH (quadratic-in-ω_H) share of the flux $\Pi(5)$ *grows monotonically* as ν decreases (from 0.5% at $\nu = 0.2$ to 23.1% at $\nu = 0.005$), i.e. the ($\star$)-consequence degrades exactly where the mechanism would be needed. The same measurements give (†-weak) *with a rate*: the low-mode dissipation share $\rho_\nu(\{|k| < 5\})/(\frac{1}{2}B_0)$ drains from 1.0 to 0.128 over $\nu \in [0.2, 0.005]$ with local decay exponents α_{loc} ($\rho_\nu \sim \nu^\alpha$) rising $0.08 \to 0.93$ — consistent with an approach to $\alpha = 1$, which would amount to the *original* uniform bound (†), not just (†-weak) (measured trend, not a claim).

(S6) Large ν: a vanishing absolute-production limit with positive limiting diagnostic. The key point is that θ is a ratio of two *quartic* functionals, hence 0-homogeneous in the field amplitude, and the rescaled OU stationary shape is ν-independent; writing $s^2 \sim B_0/\nu$ for the per-mode variance and $\varepsilon \sim s/\nu \sim \nu^{-3/2}$ for the effective coupling, the numerator $-\mathcal{C}_1 = -\nu E_\mu[Y_{\mathrm{d}}]$ has leading order $\nu \cdot s^3\varepsilon = s^4$ — the *same* order as $E_\mu[W_1]$ (Y_{d} is Gaussian-odd, so its first-order term carries ε, but the ν prefactor in $\mathcal{C}_1$ exactly restores the balance). Hence

$$\theta(\nu) \xrightarrow[\nu\to\infty]{} \theta_G(\text{shape}) \;\geq\; \theta_{G,\min} \approx 0.7948, \qquad \text{while} \qquad E_\mu[W_1],\ |\mathcal{C}_1| = O\big((B_0/\nu)^2\big) \to 0: \quad (20)$$

large viscosity is a vanishing absolute-production limit ($E[W_1]$ and the destruction both $\to 0$), not a diagnostic-null limit. Measured (exact-OU noise stepping, Appendix A): $\theta = 0.8111$–0.8123 uniformly over $\nu \in \{0.5, \dots, 20\}$, matching $\theta_G(\text{band}) = 0.8119 \pm 10^{-4}$; the direct-$\mathcal{C}_1$ route is an unreliable estimator here, with signal-to-noise $\sim \text{mean}(Y_\text{d})/\text{rms}(Y_\text{d}) \sim \varepsilon \approx 10^{-2}$ at $\nu = 10$ (measured mean/rms $\approx 1/90$; the estimator's signal $4.4 \cdot 10^{-5}$ vs block-SE $1.6 \cdot 10^{-4}$, Appendix A). Consequently the direct stationarity closure $E_\mu[I] = -\mathcal{C}_1$ is corroborated only to $\approx 5\%$ at large ν (θ via the low-variance $E_\mu[I]$ estimator is 0.812 vs θ via $-\mathcal{C}_1$ is 0.856 at $\nu = 2$, $z(\text{diff}) = -0.54$); we therefore report θ through $E_\mu[I]$ (independently corroborated by a plain Euler–Maruyama solver at 0.81216), not the noisy $-\mathcal{C}_1$ route. The parity facts themselves stand ($N[\omega]$ even, Y_d odd, $E_G[Y_\text{d}] = 0$): they say the *absolute* asymmetry vanishes as $O(\varepsilon)$, not that the ratio θ does. The outlook consequently *flips*: the FFS uniqueness/mixing regime is exactly where θ has a positive Gaussian limit, so **a rigorous perturbative proof of $\theta(\nu) \to \theta_G > 0$ at large ν — positivity of the production diagnostic in the rigorous regime — is a concrete open target** (Poisson-equation/Wick computation about the OU measure), whereas the uniform small-ν floor remains the genuine closure obstruction. At the *Galerkin* level this target is now met:

Theorem F (Large-ν limit of the diagnostic, Galerkin level)**.** *Fix a Galerkin cutoff $M \geq k_f$ and consider the spectral Galerkin system $d\omega = (P_M N[P_M\omega] + \nu\omega_{xx})\,dt + dW$ on the mean-zero modes $1 \leq |k| \leq M$, with homogeneous band noise, $B_0 = \operatorname{Tr} Q > 0$. There are $C = C(M, shape)$ and ν_0 such that* every *invariant measure μ_ν (no uniqueness assumed) satisfies, for $\nu \geq \nu_0$,*

$$\bigl|\, \theta(\mu_\nu) - \theta_G \,\bigr| \;\leq\; C\,\sqrt{B_0}\;\nu^{-3/2}, \tag{21}$$

where $\theta_G = E_G[I]/E_G[W_1]$ on the Gaussian OU law of the forcing shape. If moreover $M \geq 2k_f$, the same bound holds for the exact balance coefficient,

$$\bigl|\, \theta_M(\mu_\nu) - \theta_G \,\bigr| \;\leq\; C\,\sqrt{B_0}\;\nu^{-3/2} \qquad (M \geq 2k_f), \tag{22}$$

*because $E_G[R_M] = 0$ (Lemma 4.1) and $E_{\mu_\nu}[R_M]$ obeys the same quintic correction bound (Lemma B.5). In particular $\mathcal{C}_1 < 0$ — hypothesis (NEG), hence $\theta_M > 0$ — **holds as a theorem** for all $\nu \geq \nu_1(M, B_0, shape)$ and $M \geq 2k_f$: to our knowledge, the first proved positivity of this (grouping-dependent) balance coefficient for the stochastic gCLM at $a = -2$, at the Galerkin level and for the stated W_1-grouping. For the flat band $[1,4]$ the limit is, relative to the W_1-grouping of §5, the exact rational number*

$$\theta_G(\mathit{band}\,[1,4]) \;=\; \frac{2017}{2484} \;=\; 0.811996779\ldots \tag{23}$$

Proof (full argument in Appendix B; the exact constant is independently certified, Appendix A). *Sketch.* (i) *Moments:* by the hereditary $a = -2$ cancellation $\langle P_M N[P_M\omega], \omega\rangle = 0$ (S3), $V = \|\omega\|^2$ satisfies $\mathcal{L}V \leq -2\nu V + B_0$ and $\Gamma(V) \leq 4B_0 V$, giving $E_\mu[V^p] \leq (2p-1)!!\,(B_0/2\nu)^p$ for every invariant μ (concave-cutoff localization; this also justifies $E_\mu[\mathcal{L}F] = 0$ for every polynomial F at the Galerkin level). (ii) *Poisson equation:* the OU generator is diagonal-plus-nilpotent on polynomials, so $\varphi_F = \sum_j (-D^{-1}T)^j D^{-1}\tilde{F}$ is explicit, with a ν^{-1} gain. (iii) *Exact correction identity:* $E_\mu[F] - E_G[F] = -E_\mu[\langle N_M, D\varphi_{\tilde{F}}\rangle]$, a quintic bounded by $C\nu^{-1}s^5$ ($s^2 = B_0/2\nu$), against $E_G[W_1] = c_W s^4$, $c_W > 0$. The solver evolves this Galerkin system exactly in space (drift and state projected to E_M; first-order in time), so the §4 measurements (θ point estimates 0.8111–0.8123, CI95-inclusive $[0.8107, 0.8126]$, over $\nu \in [0.5, 20]$ vs $\theta_G = 0.81200$) are consistent with the Gaussian limit predicted by Theorem F. *Honest scope: the constant $C(M)$ grows with the cutoff; the untruncated (SPDE) statement needs uniform-in-M moments and Galerkin convergence of invariant measures, both open. Small ν is untouched: the closure obstruction stands. (The "every* *invariant measure, no uniqueness assumed" formulation is a structural feature of the proof, not a claim that non-uniqueness occurs at $\nu \geq \nu_0$: the Galerkin system is expected uniquely ergodic there, and it is the small-ν identities of §3–§5, where uniqueness is genuinely open, that the uniqueness-free formulation protects.)*

Code and data availability

All algebraic certifications and numerical measurements reported here are reproducible from an accompanying code repository, organized as follows.

- **Solver.** A pseudo-spectral (Fourier) discretization of (SG) with Euler–Maruyama time stepping and 2/3-dealiasing (`anc/`); validated on 12 independent identities (Verification ledger, Appendix A).
- **Symbolic certifications** of every algebraic identity (A0, A1, A1$''$, A2, the inertial formula $I(a)$, and the Injection Lemma), band-limited-spectral with residual $\equiv 0$ to machine precision (`anc/`).
- **Measurement scripts** producing the stationary-measure estimates of the enstrophy balance, the anomalous-cascade exponent, the broad flux maximum, and the production diagnostic θ, including the independent re-measurement of §4 on separately written code and the convergence matrix (`anc/`, `anc/`).
- **Obstruction analyses** for (†-weak) and $\theta \geq \theta_0 > 0$: the hereditary-cancellation certification, the flux-sign and parity obstruction checks, and the large-ν limit measurements with exact-OU noise stepping and the stationary-balance closure check (`anc/`, in particular `theta_large_nu_exact_ou.py` and `closure_balance_check.py`).

The repository is public at `https://github.com/rifmj/sgclm-cascade`; **the results of this version correspond to the tagged release** `v1.14`, archived at Zenodo, `doi:10.5281/zenodo.21492681`. The paper source is under `paper/` and the verification package (the CAS certificates, the solver, and the single-command `make verify`) under `anc/`; see `anc/README.md`. The model and its large-viscosity ergodic theory are due to Fujita–Fukuizumi–Sakajo (arXiv:2603.06182).

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Use of AI assistance. The construction and write-up of this paper were carried out with substantial assistance from AI tools, including for the exploratory derivations and for the multi-round internal referee process. All mathematical claims rest on the machine-checkable certificates (band-limited spectral certification, residuals 10^{-14}–10^{-19}) and the reproducible solver (12/12 self-test) documented above, not on trust in those tools. The author has verified the results and is solely accountable for them.

A Verification: certificates, solver, and reproducibility

Table 2: Trusted-computing-base map: stage → program → arithmetic model → certified output. Primary inputs are the committed campaign data (`anc/*.jsonl`); the tables and figures are derived outputs.

stage	program	arithmetic	certified output
identities (A0–A2, $D\mathcal{P}_1$, $I(a)$, B0), symbolic	`anc/certify_identities_symbolic.py`	exact $\mathbb{Q}[a]$	zero polynomials, $K \le 8$
identities, spectral	`anc/certify_*.py`, `anc/validate.py`	float64	residuals $\le 10^{-14}$
projected generator	`anc/certify_projected_generator.py`	exact rationals + float64	$I = I_M + R_M$; support lemma; M=1
θ_G, exact	`anc/certify_theta_G_exact.py`	exact rational Wick sums	2017/2484 (+ float MC cross-check)
stationary campaign	`anc/p2_campaign_v18.py`	float64; padded $L = 2N$ diagnostics	Table 1 + receipts
convergence matrix	`anc/p2_convergence_matrix.py` + campaign receipts	float64; unpadded ($\le 4 \cdot 10^{-6}$)	Table 4
analysis / fits	`anc/p2_analysis.py`	float64	deficit log-law, α_{loc}, shares

Numerical method and error control (scope). The solver is a pseudo-spectral (Fourier) Galerkin truncation on E_M with the **strict** 2/3 rule $M = \lfloor (N-1)/3 \rfloor$ ($3M \le N-1$, so no product of retained modes aliases onto a retained mode — with the non-strict $\lfloor N/3 \rfloor$ cutoff at $3 \mid N$ the boundary pair's self-product $2M$ aliases on the grid exactly onto the retained mode $-M$): the mask is applied to every factor entering the quadratic products *and* to the nonlinear result and the state, so the evolved system is the Galerkin system on E_M exactly (the state carries no energy above M). The diagnostic functionals I, I_M, R_M, W_1, $\langle \omega_{xx}, D\mathcal{P}_1 \rangle$ are evaluated on a **zero-padded** grid $L = 2N$ ($L/2 > 2M$ and $L > 4M$), so the quadratic fields $N[\omega]$, $D\mathcal{P}_1$ and the quartic quadratures are exact, alias-free evaluations of the paper's functionals; each campaign point also records the unpadded pseudo-spectral values as an aliasing receipt (Table 1). (The dedicated convergence-matrix driver behind Table 4 evaluates θ through the unpadded N-grid path; the recorded aliasing impact on θ is $\le 4 \cdot 10^{-6}$, far below the statistical bands, so the matrix comparisons are unaffected; its N=384 and $dt/2$ ($\nu = 0.01$) rows are padded receipts from the campaign pipeline.) Time stepping uses an integrating-factor scheme in which the viscous part and the additive-noise increment are advanced *exactly* in the linear variable (exact Ornstein–Uhlenbeck noise stepping, in all runs), while the nonlinear drift is advanced by an explicit first-order step; the scheme is therefore *exact in space* but *first-order in time*. Alongside each campaign point we record the two dimensionless diagnostics of the production balance: $\delta_M = |E[R_M]|/E[W_1]$ (the spatial-projection defect separating θ from θ_M, §4; the campaign table's $\theta - \theta_M$ column) and $\delta_{\text{bal}} = |E[I_M] + \mathcal{C}_1|/E[W_1]$ (the residual of the exact balance (10) in the time-discretized chain; dominated by the statistical error of the noisy $\mathcal{C}_1$ estimator: $1.7 \cdot 10^{-3}$–$1.7 \cdot 10^{-2}$ across the small-ν campaign, and unusable at large ν where the estimator's SNR collapses — $\delta_{\text{bal}} \approx 3$ at $\nu = 10$). (The tables' θ_M is computed through the exact balance from the projected pairing I_M; the direct $-\mathcal{C}_1/E[W_1]$ estimator differs from it by up to δ_{bal}.) Reported values are stationary block averages; ratio uncertainties are leave-one-block-out (block-jackknife) errors; split-half drift diagnostics ($z < 2$ on the observables). Controls: dealiased spectral-tail suppression ($\leq 9 \cdot 10^{-7}\times$ peak), the hereditary-lemma check $\Pi_{LL} \sim 10^{-19}$, and the convergence matrix of Table 4 (plotted in Figure 4). The quoted $\pm$ bands are conditional statistical (block-bootstrap) errors at fixed discretization; the convergence matrix bounds the discretization systematics separately: time-step halving at *every* campaign ν (and quartering at $\nu = 0.01$), a spatial-resolution increase ($N = 384$ vs 256) at the two smallest ν, an independent far-from-band initial condition, and integrated autocorrelation times τ_{int} for I, W_1 and their ratio (all $\lesssim 2$ time units, versus block length 50 time units — blocks are $\gg \tau_{\text{int}}$, validating the block statistics). Every matrix entry agrees with the base grid within the quoted statistical bands: at the reported precision the discretization bias is below the statistical resolution. In particular the $dt \to dt/2 \to dt/4$ sequence at $\nu = 0.01$ ($0.7759 \to 0.7754 \to 0.7760$) is non-monotone within one band, so no Richardson extrapolation is warranted and we do not quote θ beyond 10^{-4}; the quoted $\pm$ bands should be read as statistical with a first-order-in-time systematic bounded by the same scale.

Table 3: Campaign parameters. N = grid size (strict 2/3-dealiased Galerkin cutoff $M = \lfloor (N-1)/3 \rfloor$); all rows use the integrating-factor scheme with exact-OU noise increments, dt = nonlinear-drift step; burn-in and stationary window in time units; sampling every 0.5, blocks of 50 time units; forced band $[1, 4]$, $B_0 = 0.16$.

campaign	ν	N	dt	burn / window	seeds	max $\lvert z \rvert$
stationary (§4)	0.2–0.05	128	10^{-3}	$\frac{\max(100,5/\nu)}{\max(2000,20/\nu)}$	2	1.7
	0.02	192	$7.5\cdot 10^{-4}$	same	2	(all < 2)
	0.01, 0.005	256	$5\cdot 10^{-4}$	same	2	
large-ν	0.5–20	128	exact-OU	$dt/5$-stable	2	clean
exploratory (a-sweep)	0.005–0.1	128–256	10^{-3}	short	3	13–28 (drifting)

Table 4: Convergence matrix (one seed per check; base = the §4 campaign values). Every check is consistent with base within the quoted conditional statistical bands. τ_{int} in time units (sampling interval 0.5).

check	ν	N	dt	θ	base θ (pooled)	δ_M
$dt/2$	0.2	128	$5 \cdot 10^{-4}$	0.8065 ± 0.0004	0.8061	–
$dt/2$	0.1	128	$5 \cdot 10^{-4}$	0.7982 ± 0.0005	0.7978	–
$dt/2$	0.05	128	$5 \cdot 10^{-4}$	0.7892 ± 0.0005	0.7890	–
$dt/2$	0.02	192	$3.75 \cdot 10^{-4}$	0.7801 ± 0.0004	0.7808	–
$dt/2$	0.01	256	$2.5 \cdot 10^{-4}$	0.7754 ± 0.0003	0.7758	$3.4 \cdot 10^{-5}$
$dt/2$	0.005	256	$2.5 \cdot 10^{-4}$	0.7722 ± 0.0002	0.7722	–
$dt/4$	0.01	256	$1.25 \cdot 10^{-4}$	0.7760 ± 0.0003	0.7758	–
N=384	0.01	384	$5 \cdot 10^{-4}$	0.7760 ± 0.0004	0.7758	$1.4 \cdot 10^{-8}$
N=384	0.005	384	$5 \cdot 10^{-4}$	0.7722 ± 0.0002	0.7722	$5.3 \cdot 10^{-5}$
far IC	0.01	256	$5 \cdot 10^{-4}$	0.7764 ± 0.0003	0.7758	–
τ-ref (τ_{int}: I 1.99, θ 0.67)	0.2	128	$1 \cdot 10^{-3}$	0.8057 ± 0.0004	0.8061	–
τ-ref (τ_{int}: I 1.82, θ 1.24)	0.05	128	$1 \cdot 10^{-3}$	0.7890 ± 0.0005	0.7890	–

Algebra — symbolic layer (exact, zero polynomials over $\mathbb{Q}[a]$):

- `anc/certify_identities_symbolic.py`: the five load-bearing identities — (1) the a-cancellation $\langle N[\omega], \omega\rangle = (1 + a/2) \int (H\omega)\omega^2$, (2) the first variation $D\mathcal{P}_1$ against an independent symbolic field, (3) the five-term inertial closed form $I(a)$, (4) Injection Lemma with symbolic per-mode

Table 5: Resolution (M) convergence of the *spectral-localization* diagnostics at the two smallest viscosities (base $N = 256$, the §4 campaign, vs enhanced $N = 384$; exact-OU stepping, two seeds). In contrast to θ — M-stable to $< 0.02\%$ (Table 4) — these spectral quantities are resolution-sensitive: $\rho_\nu(\{|k| < 5\})$ and the flux ratios $\Pi(K)/\frac{1}{2}B_0$ shift by $\leq 3\%$ under M-refinement, and the HH share by up to $\sim 10\%$ (at $\nu = 0.01$). So the stability of the integral ratio θ does *not* by itself certify convergence of the spectral localization; the qualitative trends (UV-drain of ρ_ν, monotone growth of the HH share) are robust, but the two-digit spectral values at the smallest ν carry a few-percent resolution systematic. Data: `anc/p2_convergence_v18.jsonl`, `anc/p2_results_v18.jsonl` (N-up rows).

ν	N	$\rho_\nu(\{\lvert k\rvert < 5\})/\frac{1}{2}B_0$	$\Pi(5)/\frac{1}{2}B_0$	$\Pi(6)/\frac{1}{2}B_0$	$\Pi(8)/\frac{1}{2}B_0$	HH share of $\Pi(5)$
0.005	256	0.128	0.872	0.834	0.757	0.234
0.005	384	0.126	0.874	0.837	0.762	0.225
0.01	256	0.243	0.757	0.688	0.560	0.201
0.01	384	0.236	0.764	0.698	0.578	0.180

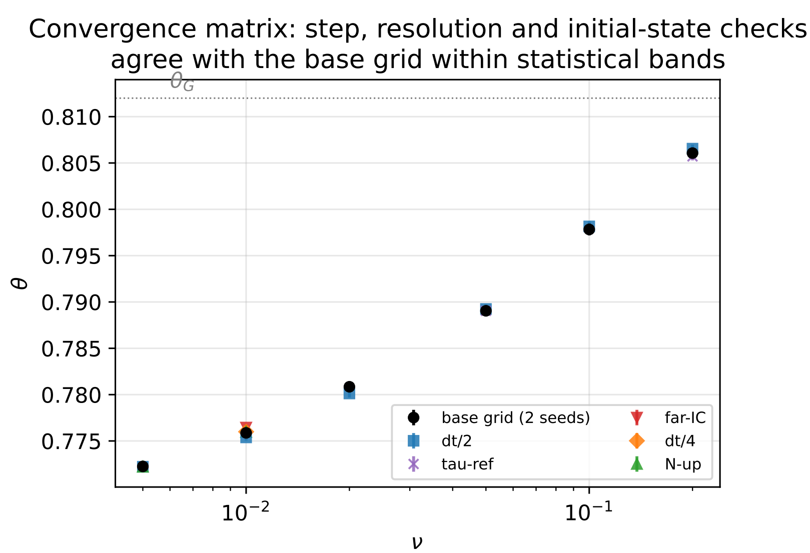

Figure 4: The convergence matrix of Table 4 plotted against the base grid: time-step halving/quartering, spatial resolution $N = 384$, and an independent far-from-band initial condition all reproduce $\theta(\nu)$ within the conditional statistical bands. Data: `anc/p2_convergence_v18.jsonl` and the tagged receipt rows (N=384; $dt/2$ at $\nu = 0.01$) of `p2_results_v18.jsonl`.

rates q_j, (5) the A2 secondary term — are each expanded to the **zero polynomial** in the symbolic mode coefficients $\{a_k, b_k\}$ (advection parameter a symbolic), for all fields band-limited to $K \le 8$. This is a proof over $\mathbb{Q}[a]$ for those bands, not a floating-point check; Appendix C proves each identity analytically for general fields, making this layer an independent verification rather than the carrier of the result.

Algebra — spectral layer (band-limited, machine precision, independent code path):

- Solver `anc/validate.py`: 12/12 ($u_x = H\omega$ $8{\cdot}10^{-17}$; $a = -2$ cancellation $4.5{\cdot}10^{-17}$; enstrophy-balance ratio 1.006; $d\mathcal{P}/dt$ chain-rule vs FD $8 \cdot 10^{-9}$).

- Flux-law layer `anc/certify_flux_law.py`: L0/L1 decompositions, 5 seeds, $\le 10^{-14}$ ($a = -2$ cancellations to 10^{-19}).

- Production-balance inertial layer `anc/certify_palinstrophy_mine.py`: $D\mathcal{P}_1$ and $I(a)$, 4 seeds, machine precision (independently re-derived on separately written code).

- Injection `anc/certify_ito_vanishing.py`: homogeneous Itô trace 10^{-16}, anisotropic $\neq 0$.

- Projected generator `anc/certify_projected_generator.py`: the decomposition $I = I_M + R_M$; the support lemma ($R_M = 0$ for fields with $2K \le M$, exact zero) and its sharpness ($R_M \neq 0$ for $M < 2K$); the $M{=}1$ example of Remark 4.2 to machine precision against the exact rationals $I = \frac{15\pi}{4}A^4$, $W_1 = \frac{9\pi}{2}A^4$, $I_M = \mathcal{C}_1 = 0$.

- Exact Gaussian constant `anc/certify_theta_G_exact.py`: the five Wick sums in rational arithmetic, $\theta_G(\text{band}[1,4]) = 2017/2484$; scale-invariance and reality checks pass; MC cross-check through the independent solver-side code path to $5.6 \cdot 10^{-5}$; the exact rational is independently re-derived by pointwise Wick moments in Appendix B.

Measurement (dynamical solver at stationarity; the §4 stationary campaign unless noted):

- (L0) $\nu E[\int \omega_x^2]/(\frac{1}{2}B_0) \in [0.977, 1.030]$ (seed-pooled per ν; per-seed $[0.95, 1.06]$) across $\nu \in [0.005, 0.2]$ (uniform).

- Anomalous cascade $E[\mathcal{E}_1] \sim \nu^{-1.002}$, $R^2 = 0.9999$, prefactor 0.0400 vs $B_0/4 = 0.0400$.

- (FL∞) broad flux maximum (not a resolved plateau), through the Theorem A1 identity $\Pi(K) = \frac{1}{2}B_0 - \rho_\nu(\{|k| < K\})$ ($K > k_f$): $\Pi/(\frac{1}{2}B_0) = 0.872, 0.834, 0.757, 0.600$ at $K = 5, 6, 8, 12$, $\nu = 0.005$ (maximum approaching 1 as $\nu \to 0$; <1 octave of scale separation).

- a-monotonicity: $E[\mathcal{P}_1]$ rises monotonically ($\approx 3\times$) over $a : -2 \to -0.5$, blow-up at $a = 0$ (exploratory sweep; qualitative, no error bars quoted).

- Large-ν limit (exact-OU stepping): $\theta_G(\text{band}) = 0.8119 \pm 10^{-4}$ (Gaussian sampling, $N_{\mathrm{MC}} = 2 \cdot 10^5$); SDE grid $\nu \in \{0.5, 1, 2, 5, 10, 20\} \times 2$ seeds, all $\theta \in [0.8107, 0.8126]$ including CI95 half-widths (point estimates 0.8111–0.8123; block-bootstrap CI95, split-half z clean, $dt/5$-stable); balance closure at $\nu = 2$: $E[I] = 1.1124 \cdot 10^{-3} \pm 8 \cdot 10^{-6}$ vs $-\nu E[Y] = 1.1730 \cdot 10^{-3} \pm 1.1 \cdot 10^{-4}$, $z(\text{diff}) = -0.54$ (`anc/theta_large_nu_exact_ou.log`, `anc/closure_balance_check.log`).

- Stationary campaign (§4 table; `anc/p2_campaign_v18.py` + `p2_results_v18.jsonl` + `p2_analysis.py`): 6 ν-points × 2 seeds, split-half max $|z| = 1.7$ (all < 2) on $I, W_1, \mathcal{E}_1, D$; deficit log-law, ρ_ν rates, ($\star$) refutation, and the moment-form Cor E inputs ($E[Y^2]$, negative inertial-generation fraction 0) all from the same runs.

- Convergence matrix (Table 4; `anc/p2_convergence_matrix.py (driven on the projected solver) + p2_convergence_v18.jsonl`): $dt/2$ at every campaign ν, $dt/4$ at $\nu = 0.01$, $N = 384$ at $\nu \in \{0.01, 0.005\}$, an independent far-from-band initial condition, and τ_{int} for I, W_1, θ — all consistent with the base grid.

- *The coarse exploratory sweep (stationarity-z 13–28; $\nu = 0.02$ under-resolved; 3-seed ensembles) is retained only for the qualitative a-dependence; every quantitative flux/scaling number above is from the stationary campaign.*

Cross-validation (independent implementations): the inertial formula $I(a)$ is certified independently in `anc/certify_palinstrophy_mine.py` and by a separate, independently-coded three-derivation of $D\mathcal{P}_1$ from the solver primitives (each written blind to the other); the two agree to machine precision.

B Proof of Theorem F (large-ν limit of the diagnostic, Galerkin level)

Fix the Galerkin cutoff $M \geq k_f$ and work on $E_M = P_M L^2_0(\mathbb{T})$ (real, mean-zero, modes $1 \leq |k| \leq M$), with

$$d\omega = \big(N_M[\omega] + \nu\omega_{xx}\big)\,dt + dW, \qquad N_M[\omega] = P_M N[P_M\omega], \tag{24}$$

$N[\omega] = -au\omega_x + v\omega$ at $a = -2$, homogeneous band noise $b_k = f(|k|)$ on $1 \leq |k| \leq k_f \leq M$, and $B_0 = \sum_k b_k^2 = \operatorname{Tr} Q > 0$. Write $V(\omega) = \|\omega\|^2_{L^2}$ and $s^2 = B_0/(2\nu)$. On the finite-dimensional E_M all Sobolev norms are equivalent (constants depending on M), and I, W_1 and the Poisson potentials below are polynomials of degree ≤ 5 in the mode coordinates. The Itô generator splits as $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1$, with $\mathcal{L}_0 F = \langle \nu\omega_{xx}, DF\rangle + \frac{1}{2}\operatorname{Tr}(QD^2F)$ (the Ornstein–Uhlenbeck part) and $\mathcal{L}_1 F = \langle N_M[\omega], DF\rangle$.

The Gaussian reference measure, explicitly. The stationary law of $\mathcal{L}_0$ is the **degenerate** Gaussian

$$\begin{aligned} G_\nu \;=\; \mathcal{N}\Big(0,\ \operatorname{diag}\big(b_k^2/(2\nu k^2)\big)\Big) &\text{ on the forced band } 1 \leq |k| \leq k_f \\ \otimes\ \delta_0 &\text{ on the unforced modes } k_f < |k| \leq M: \end{aligned} \tag{25}$$

the OU drift $-\nu k^2$ damps every mode, but noise enters only the band, so the unforced modes relax to zero under $\mathcal{L}_0$. All Wick sums below therefore range over the band modes only (this is why θ_G is a functional of the forcing *shape*). Note the invariant measures μ_ν of the full *nonlinear* system are not supported on the band: $\mathcal{L}_1$ transfers mass to all modes $\leq M$; only the reference Gaussian is degenerate.

Norms. For a polynomial $F = \sum_\alpha c_\alpha x^\alpha$ in the mode coordinates $\{x_k\}$ of E_M, write $\|F\|_{\text{coeff}} := \max_\alpha |c_\alpha|$ (maximal coefficient magnitude in the monomial basis); this is the norm used for the coefficient bounds below. Throughout, $\theta_G = E_{G_\nu}[I]/E_{G_\nu}[W_1]$, which is ν-independent (both quartic, the rescaled shape ν-independent).

Lemma B.1 (Hereditary cancellation)**.** $\langle N_M[\omega], \omega\rangle = 0$ *for every* $\omega \in E_M$.

Proof. As $\omega \in E_M$ is band-limited to $|k| \le M$, the idempotent P_M is self-adjoint and fixes ω, so $\langle N_M[\omega], \omega\rangle = \langle N[\omega], \omega\rangle = (1 + a/2) \int (H\omega)\omega^2 = 0$ at $a = -2$ (Theorem A0). $\square$

Lemma B.2 (Moment bounds)**.** *For every $p \ge 1$ and every invariant measure μ_ν of the Galerkin system, $E_{\mu_\nu}[V^p] \le (2p-1)!!\,(B_0/2\nu)^p = (2p-1)!!\,s^{2p}$; consequently $E_{\mu_\nu}[\mathcal{L}F] = 0$ for every polynomial F.*

Proof. By Lemma B.1, $\mathcal{L}V = -2\nu\|\omega_x\|^2 + B_0 \le -2\nu V + B_0$ (Poincaré $\|\omega_x\|^2 \ge \|\omega\|^2$ on mean-zero modes; the Itô trace is B_0), and the carré du champ satisfies $\Gamma(V) = 4\sum_{k\in\text{band}} b_k^2|\hat\omega_k|^2 \le 4B_0V$. Hence

$$\mathcal{L}V^p = pV^{p-1}\mathcal{L}V + \tfrac{1}{2}p(p-1)V^{p-2}\Gamma(V) \le -2p\nu V^p + p(2p-1)B_0V^{p-1}. \tag{26}$$

To use $E_\mu[\mathcal{L}(\cdot)] = 0$ before finiteness of moments is known, apply the generator to concave cutoffs $\chi_R(V^p)$ (χ_R smooth, concave, $\chi_R(t) = t$ for $t \le R$, $\chi_R' \in [0,1]$, $\chi_R'' \le 0$, $\chi_R \uparrow \mathrm{id}$): then $\mathcal{L}\chi_R(V^p) = \chi_R'\,\mathcal{L}V^p + \frac{1}{2}\chi_R''\,\Gamma(V^p) \le \chi_R'\,\mathcal{L}V^p$ ($\Gamma \ge 0$, $\chi_R'' \le 0$), and $\chi_R(V^p)$ is bounded with bounded derivatives, so $E_\mu[\mathcal{L}\chi_R(V^p)] = 0$ for every invariant μ. Proceed by induction on p (base $E_\mu[V^0] = 1$), assuming $E_\mu[V^{p-1}] < \infty$. Combining the two displays,

$$\begin{aligned} 0 \;&\le\; E_\mu\big[\chi_R'(V^p)\big(-2p\nu V^p + p(2p-1)B_0V^{p-1}\big)\big], \\ \text{i.e.}\qquad 2p\nu\, E_\mu[\chi_R'(V^p)\,V^p] \;&\le\; p(2p-1)B_0\, E_\mu[\chi_R'(V^p)\,V^{p-1}], \end{aligned}$$

where the right side is $\le p(2p-1)B_0\,E_\mu[V^{p-1}] < \infty$ ($0 \le \chi_R' \le 1$, $V \ge 0$) — note both integrands are nonnegative, so no sign-splitting is needed. As $R \uparrow \infty$, $\chi_R'(V^p)\,V^p \to V^p$ pointwise, and Fatou's lemma gives $2p\nu\,E_\mu[V^p] \le p(2p-1)B_0\,E_\mu[V^{p-1}]$ — in particular $E_\mu[V^p] < \infty$ — i.e. $E_\mu[V^p] \le (2p-1)(B_0/2\nu)\,E_\mu[V^{p-1}]$, and the induction yields the bound. With all moments finite, dominated localization gives $E_\mu[\mathcal{L}F] = 0$ for every polynomial F, using neither uniqueness nor ergodicity. $\square$

For a polynomial P homogeneous of degree d, the bound and Hölder give $E_{\mu_\nu}|P| \le C(P,M)\,s^d$.

Lemma B.3 (Poisson equation on polynomials)**.** *For every polynomial F of degree d with $E_{G_\nu}[F] = 0$ there is a polynomial φ_F of the same degree with $\mathcal{L}_0\varphi_F = F$ and, writing $\varphi_F = \sum_q \varphi_F^{(q)}$ for its degree-q homogeneous parts, the coefficient bounds $\|\varphi_F^{(q)}\|_{\text{coeff}} \le C(F,M)\,\nu^{-1}\max\!\big(1, (B_0/\nu)^{(d-q)/2}\big)$, where $C(F,M)$ depends only on $\|F\|_{\text{coeff}}$, d, and the number of modes.*

Proof. In mode coordinates $\mathcal{L}_0 = D + T$, where $D = -\sum_k \gamma_k x_k \partial_{x_k}$ ($\gamma_k = \nu k^2$) is diagonal on monomials with eigenvalue $-\sum_k \alpha_k\gamma_k \le -\nu$ on every non-constant monomial (since $\sum_k \alpha_k k^2 \ge 1$) and kills constants, and $T = \frac{1}{2}\sum_{k\in\text{band}} q_k\partial_{x_k}^2$ lowers degree by two ($q_k \le B_0$). Let Π_0 be the projection onto the constant monomial and define D^{-1} *on* $\mathrm{Ran}(1-\Pi_0)$ *only*, where D is invertible with $\|D^{-1}\| \le \nu^{-1}$ (a centered polynomial can carry a nonzero constant *monomial* even though its Gaussian mean vanishes, so this restriction is not vacuous). Set $\varphi_F = \sum_{j\ge0}\big(-D^{-1}(1-\Pi_0)T\big)^j D^{-1}(1-\Pi_0)F$ (a finite sum, $j \le d/2$, since T lowers degree): dropping the constant component at each stage yields $\mathcal{L}_0\varphi_F = F + c$ for some constant c. Taking E_{G_ν} of both sides: $E_{G_\nu}[\mathcal{L}_0\varphi_F] = 0$ (G_ν is the stationary law of $\mathcal{L}_0$) and $E_{G_\nu}[F] = 0$, so $c = 0$. Each D^{-1} contributes a factor $\le \nu^{-1}$, each T a factor $\le B_0$. $\square$

Lemma B.4 (Correction identity)**.** *For every polynomial F and every invariant μ_ν, with $\tilde F = F - E_{G_\nu}[F]$,*

$$E_{\mu_\nu}[F] - E_{G_\nu}[F] = -\,E_{\mu_\nu}\big[\langle N_M[\omega], D\varphi_{\tilde F}\rangle\big]. \tag{27}$$

Proof. $0 = E_\mu[\mathcal{L}\varphi_{\tilde F}] = E_\mu[\mathcal{L}_0\varphi_{\tilde F}] + E_\mu[\mathcal{L}_1\varphi_{\tilde F}] = E_\mu[\tilde F] + E_\mu[\langle N_M, D\varphi_{\tilde F}\rangle]$, the generator identity for the polynomial $\varphi_{\tilde F}$ being justified by Lemma B.2. $\square$

Lemma B.5 (Galerkin-defect bound). *Let $M \geq 2k_f$. Then $E_{G_\nu}[R_M] = 0$, and for every invariant measure μ_ν of the Galerkin system, $|E_{\mu_\nu}[R_M]| \leq C(M)\,\nu^{-1}s^5$.*

Proof. G_ν is supported on the forced band $|k| \leq k_f$ (the degenerate Gaussian above), and on band-supported fields R_M vanishes pointwise by Lemma 4.1 (its two factors are quadratic, hence supported on modes $\leq 2k_f \leq M$, which Q_M annihilates); so $E_{G_\nu}[R_M] = 0$. Since R_M is a quartic polynomial in the mode coordinates of E_M, Lemma B.4 with the coefficient bounds of Lemma B.3 gives the quintic bound $|E_{\mu_\nu}[R_M] - E_{G_\nu}[R_M]| \leq C(M)\,\nu^{-1}s^5$, exactly as for $F \in \{I, W_1\}$ in the assembly below. $\square$

Proof of Theorem F. Apply Lemma B.4 to $F \in \{I, W_1\}$ (quartic, $d = 4$). The integrand $\langle N_M[\omega], D\varphi_{\tilde{F}}\rangle$ is a polynomial whose degree-$(5-2j)$ part carries coefficients $\leq C(M,F)\,B_0^j\nu^{-j-1}$ in $\|\cdot\|_{\mathrm{coeff}}$ (Lemma B.3; N_M is quadratic with M-bounded coefficient tensors). Here and below, $C(M,F)$ aggregates three M-dependent quantities: the number of monomials of the quartic F (at most $O(M^4)$), $\max_{k,l}$ of the quadratic-form coefficients of N_M (polynomial in M through the u, v, ∂_x multipliers on modes $\leq M$), and the length $\leq d/2+1$ of the nilpotent series in Lemma B.3 — this is the sole source of the M-growth of the constant in Theorem F. By the consequence of Lemma B.2 and $s^2 = B_0/2\nu$,

$$|E_\mu[F] - E_G[F]| \;\leq\; \sum_{j=0}^{2} C\,B_0^j\,\nu^{-j-1}\,s^{5-2j} \;=\; C\,\nu^{-1}s^5\sum_{j=0}^{2}\Big(\tfrac{B_0}{\nu s^2}\Big)^j \;=\; C'\,\nu^{-1}s^5, \tag{28}$$

the sum being finite ($B_0/(\nu s^2) = 2$ and j ranges over $\{0,1,2\}$). Both Gaussian means are exactly quartic, $E_G[W_1] = c_W s^4$ and $E_G[I] = c_I s^4$ with $c_W > 0$ (as $W_1 \geq 0$ and $E_G[W_1] = 0$ only if $v\omega_x \equiv 0$ G-a.s., impossible for a non-degenerate band Gaussian). Hence, for $\nu \geq \nu_0$ so that $E_\mu[W_1] \geq c_W s^4/2$,

$$|\theta(\mu_\nu) - \theta_G| = \frac{|E_\mu I \cdot E_G W_1 - E_G I \cdot E_\mu W_1|}{E_\mu W_1 \cdot E_G W_1} \;\leq\; \frac{C\,(\nu^{-1}s^5)\,s^4}{s^8} \;=\; C\,\frac{s}{\nu} \;=\; C\sqrt{B_0/2}\,\nu^{-3/2}. \tag{29}$$

For the balance coefficient at $M \geq 2k_f$: by Lemma B.5 and $E_\mu[W_1] \geq c_W s^4/2$,

$$|\theta_M(\mu_\nu) - \theta(\mu_\nu)| = \frac{|E_\mu[R_M]|}{E_\mu[W_1]} \leq \frac{C\,\nu^{-1}s^5}{c_W s^4/2} = C'\sqrt{B_0/2}\,\nu^{-3/2}, \tag{30}$$

and the θ_M bound follows by the triangle inequality; since $\theta_G \geq \theta_{G,\min} > 0$ (Lemma B.6), $\mathcal{C}_1 = -\theta_M(\mu_\nu)\,E_\mu[W_1] < 0$ for all $\nu \geq \nu_1(M, B_0, \text{shape})$. $\square$

Lemma B.6 (Positivity of the Gaussian limit). *For homogeneous band noise with arbitrary amplitudes $b_k \geq 0$ (not all zero) on $1 \leq |k| \leq k_f$,*

$$\theta_G \;=\; \frac{E_G[I]}{E_G[W_1]} \;\geq\; \theta_{G,\min} \;=\; \frac{1}{3} + \frac{\sqrt{69}}{18} \;>\; 0.$$

Proof. Wick's theorem over the band Gaussian (per-pair variances s_k^2) gives

$$E_G[W_1] = \sum_k 36\pi k^2 s_k^4 + \sum_{i<j} 12\pi\,(i^2 + 4ij + j^2)\,s_i^2 s_j^2,$$

a direct two-mode computation (the weights of the modes actually forced are strictly positive, while zero-amplitude modes contribute zero-weight terms that drop from the sums; at least one positive term survives since not all b_k vanish, so $E_G[W_1] > 0$), and correspondingly $E_G[I] = \sum_k \frac{5}{6} \cdot 36\pi k^2 s_k^4 + \sum_{i<j} \theta_G(j/i) \cdot 12\pi(i^2+4ij+j^2)\, s_i^2 s_j^2$, where $\frac{5}{6}$ is the certified diagonal self-ratio and $\theta_G(m) = (5+7m+3m^2)/(3+12m+3m^2)$ the certified pairwise ratio (§4). Every ratio appearing is $\geq \min\big(\frac{5}{6},\ \min_{m>1}\theta_G(m)\big) = \theta_{G,\min}$ (interior minimum at $m^* = (2+\sqrt{69})/5$, §4). For positive weights $\beta_\alpha > 0$ and ratios $r_\alpha \geq r$, the weighted mediant satisfies $\sum_\alpha \beta_\alpha r_\alpha / \sum_\alpha \beta_\alpha \geq r$; apply it with $\beta =$ the displayed W_1-weights times the variance monomials. $\square$

The exact constant. For the flat band $[1,4]$ ($\sigma_k^2 \propto 1/k^2$) the five Wick sums evaluate in exact rational arithmetic (`anc/certify_theta_G_exact.py`, Monte-Carlo cross-checked through the independent solver-side code path to $5.6\cdot 10^{-5}$) to $T_1 = 115/2$, $T_2 = T_3 = T_5 = -625/36$, $T_4 = -395/36$ (in units $2\pi\cdot\mathrm{scale}^2$, with per-pair variances $s_k^2 = \mathrm{scale}^2/k^2$ and $\mathrm{scale}^2 = 2$ for the tabulated values; only the final ratio is normalization-independent), giving $E_G[I] = (10085/36)\cdot 2\pi$, $E_G[W_1] = 345\cdot 2\pi$, and

$$\theta_G(\text{band}\,[1,4]) = \frac{10085/36}{345} = \frac{2017}{2484} = 0.811996779\ldots \tag{31}$$

A pointwise-moment cross-derivation of T_1: by Gaussian factorization $E_G[T_1]/2\pi = E[v^2]\,E[\omega_x^2] + 2\,E[v\omega_x]^2 = (205/72)\cdot 8 + 2(25/6)^2 = 115/2$; the coincidence $T_2 = T_3 = T_5 = -625/36$ is structural, and for T_5 the internal pairing vanishes on $\mathrm{sgn}(0) = 0$. The value $\theta_{G,\min} = 1/3 + \sqrt{69}/18 \approx 0.7948$ is the infimum of the *cross-mode coefficient ratio* $\theta_G(m)$ — a universal lower bound, not an attained full-shape minimum; the band value sits strictly above it, being the c_W-weighted mediant of the diagonal self-ratio 5/6 and the cross ratios, all $\geq \theta_{G,\min}$ with positive weights.

Honest scope. The constant $C = C(M)$ grows with the cutoff (through the norm-equivalence constants and the coefficient tensors of N_M); removing it needs uniform-in-M moment bounds in a norm strong enough for the quintic error together with Galerkin convergence of invariant measures — both open. The small-ν regime is untouched.

C Proofs of the algebraic identities (general fields)

This appendix proves every load-bearing algebraic identity of §3–§4 for general smooth mean-zero fields on $\mathbb{T}$ — in particular for trigonometric polynomials of *arbitrary* band — so the symbolic $K \leq 8$ certificates of Appendix A are an independent verification layer, not the carrier of the results. We use three standard properties of the Hilbert transform on mean-zero fields:

(H1) skew-adjointness: $\int (Hf)\, g = -\int f\,(Hg)$;

(H2) commutation with derivatives: $H\partial_x = \partial_x H$ (both are Fourier multipliers); in particular $v_x = H\omega_x$ for $v = H\omega$;

(H3) $u = -\Lambda^{-1}\omega$, so $u_x = H\omega = v$.

Every identity below is an exact integral identity in the field; each is bilinear or multilinear in ω, so it holds identically for $\omega \in E_M$ at any Galerkin cutoff M (where needed, P_M is self-adjoint and fixes its range, so $\langle P_M N[P_M\omega], g\rangle = \langle N[\omega], P_M g\rangle = \langle N[\omega], g\rangle$ **for band-limited** $g \in E_M$). *This caveat is load-bearing:* in the stationarity balances the role of g is played by the test function DF, which is band-limited for the §3 balances ($g = \omega,\ P_{<K}\omega,\ \omega_{xx} \in E_M$) but **not** for the production balance of §4, whose gradient $D\mathcal{P}_1$ is quadratic and leaves E_M — there the exact Galerkin statement uses the projected pairing $I_M = \langle P_M N, P_M D\mathcal{P}_1\rangle$ with the defect R_M (§4, Lemma 4.1). The generator identities $E_\mu[\mathcal{L}F] = 0$ invoked below are theorems at any fixed Galerkin truncation (Lemma B.2), with the gradient read as $P_M DF$, and formal for the untruncated SPDE (§2 convention).

C.1 The advective cancellation

Lemma C.1 (Advective cancellation)**.** *For every smooth mean-zero ω and every a,*

$$\langle N[\omega], \omega\rangle \;=\; \Big(1+\frac{a}{2}\Big)\int (H\omega)\,\omega^2, \qquad N[\omega] = -au\omega_x + v\omega. \tag{32}$$

In particular $\langle N[\omega], \omega\rangle = 0$ at $a = -2$.

Proof. $\langle v\omega, \omega\rangle = \int v\,\omega^2$. For the transport term, integrate by parts using (H3):

$$\langle -au\omega_x, \omega\rangle = -a\int u\,\omega\,\omega_x = -\frac{a}{2}\int u\,(\omega^2)_x = \frac{a}{2}\int u_x\,\omega^2 = \frac{a}{2}\int v\,\omega^2. \tag{33}$$

Adding the two terms gives the claim. □

C.2 Stationarity balances: proofs of Theorems A0 and A1

Fix K and set $F_K(\omega) = \frac{1}{2}\|P_{<K}\omega\|_{L^2}^2$. Then $DF_K = P_{<K}\omega$ and $D^2F_K = P_{<K}$, so

$$\mathcal{L}F_K = \langle P_{<K}\omega,\, N[\omega]\rangle + \nu\,\langle P_{<K}\omega,\, \omega_{xx}\rangle + \tfrac{1}{2}\operatorname{Tr}(QP_{<K}). \tag{34}$$

Since $P_{<K}$ is a self-adjoint idempotent commuting with ∂_x: $\langle P_{<K}\omega, N[\omega]\rangle = \langle P_{<K}\omega, P_{<K}N[\omega]\rangle = -\Pi(K)$ (the definition of the flux, §2); $\nu\langle P_{<K}\omega, \omega_{xx}\rangle = -\nu\|P_{<K}\omega_x\|^2$ (integration by parts); and $\operatorname{Tr}(QP_{<K}) = \sum_{|k|<K} b_k^2 = B_0^{<K}$ (the basis $\{e_k\}$ diagonalizes both Q and $P_{<K}$). Taking $E_\mu[\mathcal{L}F_K] = 0$,

$$E_\mu[\Pi(K)] = \tfrac{1}{2}B_0^{<K} - \nu\, E_\mu\|P_{<K}\omega_x\|^2, \tag{35}$$

which is Theorem A1. Letting $P_{<K} = \mathrm{Id}$ (i.e. K beyond the cutoff, or $K\to\infty$), the flux term is $\langle \omega, N[\omega]\rangle = 0$ at $a=-2$ by Lemma C.1, and the identity becomes $\nu E_\mu[\int \omega_x^2] = \frac{1}{2}B_0$ — Theorem A0. □

C.3 Proof of Theorem A2

Apply the generator to the palinstrophy $\mathcal{E}_1 = \frac{1}{2}\int\omega_x^2$: $D\mathcal{E}_1 = -\omega_{xx}$ and $D^2\mathcal{E}_1(h,h) = \|h_x\|^2$, so $\operatorname{Tr}(QD^2\mathcal{E}_1) = \sum_k b_k^2\,\|(e_k)_x\|^2 = \sum_k b_k^2 k^2 = B_2$ (the $\{e_k\}$ are L^2-normalized). Hence $E_\mu[\mathcal{L}\mathcal{E}_1] = 0$ reads

$$\nu\, E_\mu\Big[\int \omega_{xx}^2\Big] = -\,E_\mu\big[\langle \omega_{xx}, N[\omega]\rangle\big] + \tfrac{1}{2}B_2. \tag{36}$$

It remains to identify the nonlinear term. Using (H3) and integration by parts,

$$\begin{aligned}
-\langle \omega_{xx}, N[\omega]\rangle &= a\int u\,\omega_x\omega_{xx} - \int v\,\omega\,\omega_{xx}\\
&= \frac{a}{2}\int u\,(\omega_x^2)_x + \int (v\omega)_x\,\omega_x = -\frac{a}{2}\int v\,\omega_x^2 + \int v_x\,\omega\,\omega_x + \int v\,\omega_x^2\\
&= \Big(1-\frac{a}{2}\Big)\int v\,\omega_x^2 + \int v_x\,\omega\,\omega_x.
\end{aligned} \tag{37}$$

At $a=-2$ this is $2\int(H\omega)\omega_x^2 + \int (H\omega_x)\,\omega\,\omega_x = \Pi_1$. For the secondary term, by (H2) and integration by parts,

$$\int (H\omega_x)\,\omega\,\omega_x = \frac{1}{2}\int (H\omega_x)\,(\omega^2)_x = -\frac{1}{2}\int (H\omega_x)_x\,\omega^2 = -\frac{1}{2}\int (H\omega_{xx})\,\omega^2. \qquad \square \tag{38}$$

C.4 The first variation of $\mathcal{P}_1$

Lemma C.2 (First variation)**.** *For $\mathcal{P}_1(\omega) = \int (H\omega)\,\omega_x^2$,*

$$D\mathcal{P}_1 = -H(\omega_x^2) - 2\,(H\omega_x)\,\omega_x - 2\,(H\omega)\,\omega_{xx}. \tag{39}$$

Proof. For a smooth perturbation h,

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \mathcal{P}_1(\omega+\varepsilon h) = \int (Hh)\,\omega_x^2 + 2\int (H\omega)\,\omega_x\,h_x. \tag{40}$$

By (H1) the first term is $-\int h\,H(\omega_x^2)$. Integrating the second by parts and using (H2),

$$2\int (H\omega)\,\omega_x\,h_x = -2\int \big((H\omega)\,\omega_x\big)_x h = -2\int (H\omega_x)\,\omega_x\,h - 2\int (H\omega)\,\omega_{xx}\,h. \tag{41}$$

Collecting the kernels gives the displayed $D\mathcal{P}_1$. $\square$

C.5 The inertial closed form $I(a)$ and the W_1–N_1 split

Proposition C.3 (Inertial closed form)**.** *With $b_{nl} = -au\omega_x + v\omega$ and $D\mathcal{P}_1$ as in Lemma C.2,*

$$\begin{aligned} I(a) := \langle b_{nl}, D\mathcal{P}_1\rangle = (2-2a)\int v^2\omega_x^2 - 2a\int uv\,\omega_x\omega_{xx} + 2\int vv_x\,\omega\omega_x \\ + a\int u\,\omega_x\,H(\omega_x^2) - \int v\,\omega\,H(\omega_x^2). \end{aligned} \tag{42}$$

At $a=-2$,

$$I(-2) = \underbrace{6\int v^2\omega_x^2}_{=W_1} + 4\int uv\,\omega_x\omega_{xx} + 2\int vv_x\,\omega\omega_x - 2\int u\,\omega_x H(\omega_x^2) - \int v\,\omega\,H(\omega_x^2), \tag{43}$$

so the nonlocal term in the split $I = W_1 - N_1$ is, explicitly,

$$N_1 = -4\int uv\,\omega_x\omega_{xx} - 2\int vv_x\,\omega\omega_x + 2\int u\,\omega_x H(\omega_x^2) + \int v\,\omega\,H(\omega_x^2). \tag{44}$$

Proof. Expanding $\langle -au\omega_x + v\omega,\ D\mathcal{P}_1\rangle$ into the six pairings (using $v_x = H\omega_x$, (H2)):

$$\begin{aligned} \langle v\omega, D\mathcal{P}_1\rangle &= -\int v\omega\,H(\omega_x^2) - 2\int vv_x\,\omega\omega_x - 2\int v^2\,\omega\omega_{xx}, \\ \langle -au\omega_x, D\mathcal{P}_1\rangle &= a\int u\,\omega_x H(\omega_x^2) + 2a\int uv_x\,\omega_x^2 + 2a\int uv\,\omega_x\omega_{xx}. \end{aligned} \tag{45}$$

Two integration-by-parts conversions bring this to the displayed form:

(i) $-2\int v^2\,\omega\omega_{xx} = 2\int (v^2\omega)_x\,\omega_x = 4\int vv_x\,\omega\omega_x + 2\int v^2\omega_x^2$;

(ii) $0 = \int (uv\,\omega_x^2)_x = \int v^2\omega_x^2 + \int uv_x\,\omega_x^2 + 2\int uv\,\omega_x\omega_{xx}$ (using $u_x = v$), whence $\int uv_x\,\omega_x^2 = -\int v^2\omega_x^2 - 2\int uv\,\omega_x\omega_{xx}$.

Substituting (i) and (ii),

$$I(a) = (2-2a)\int v^2\omega_x^2 + 2\int vv_x\,\omega\omega_x - \int v\omega\, H(\omega_x^2) + a\int u\,\omega_x H(\omega_x^2) - 2a\int uv\,\omega_x\omega_{xx}, \quad (46)$$

which is the claim; the $a=-2$ specialization and the definition $N_1 := W_1 - I$ give the split. $\square$

All five identities of this appendix are certified independently — symbolically as zero polynomials over $\mathbb{Q}[a]$ for bands $K \le 8$ and spectrally at machine precision (Appendix A); the derivations above remove the band restriction.

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